

Chapter 5

Lightwave Systems

The preceding three chapters focused on the three main components of a fiber-optic communication system—optical fibers, optical transmitters, and optical receivers. In this chapter we consider the issues related to system design and performance when the three components are put together to form a practical lightwave system. Section 5.1 provides an overview of various system architectures. The design guidelines for fiber-optic communication systems are discussed in Section 5.2 by considering the effects of fiber losses and group-velocity dispersion. The power and the rise-time budgets are also described in this section. Section 5.3 focuses on long-haul systems for which the nonlinear effects become quite important. This section also covers various terrestrial and undersea lightwave systems that have been developed since 1980. Issues related to system performance are treated in Section 5.4 with emphasis on performance degradation occurring as a result of signal transmission through the optical fiber. In Section 5.5 we emphasize the importance of computer-aided design for lightwave systems.

5.1 System Architectures

From an architectural standpoint, fiber-optic communication systems can be classified into three broad categories—point-to-point links, distribution networks, and local-area networks [1]–[9]. This section focuses on the main characteristics of these three system architectures.

5.1.1 Point-to-Point Links

Point-to-point links constitute the simplest kind of lightwave systems. Their role is to transport information, available in the form of a digital bit stream, from one place to another as accurately as possible. The link length can vary from less than a kilometer (short haul) to thousands of kilometers (long haul), depending on the specific application. For example, optical data links are used to connect computers and terminals within the same building or between two buildings with a relatively short transmission distance (<10 km). The low loss and the wide bandwidth of optical fibers are not of

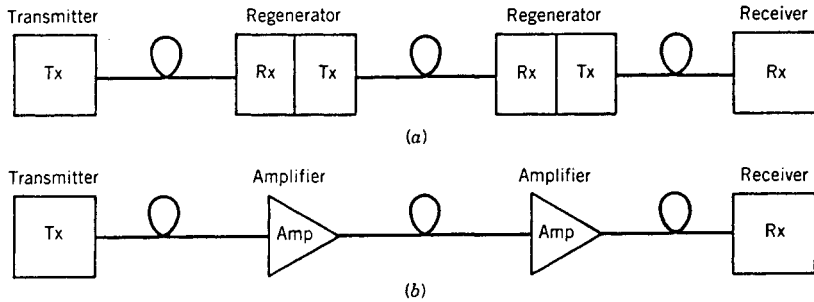


Figure 5.1: Point-to-point fiber links with periodic loss compensation through (a) regenerators and (b) optical amplifiers. A regenerator consists of a receiver followed by a transmitter.

primary importance for such data links; fibers are used mainly because of their other advantages, such as immunity to electromagnetic interference. In contrast, undersea lightwave systems are used for high-speed transmission across continents with a link length of several thousands of kilometers. Low losses and a large bandwidth of optical fibers are important factors in the design of transoceanic systems from the standpoint of reducing the overall operating cost.

When the link length exceeds a certain value, in the range 20–100 km depending on the operating wavelength, it becomes necessary to compensate for fiber losses, as the signal would otherwise become too weak to be detected reliably. Figure 5.1 shows two schemes used commonly for loss compensation. Until 1990, optoelectronic repeaters, called *regenerators* because they regenerate the optical signal, were used exclusively. As seen in Figure 5.1(a), a regenerator is nothing but a receiver–transmitter pair that detects the incoming optical signal, recovers the electrical bit stream, and then converts it back into optical form by modulating an optical source. Fiber losses can also be compensated by using optical amplifiers, which amplify the optical bit stream directly without requiring conversion of the signal to the electric domain. The advent of optical amplifiers around 1990 revolutionized the development of fiber-optic communication systems. Such amplifiers are especially valuable for WDM systems (see Chapter 6) as they can amplify a large number of channels simultaneously.

Optical amplifiers solve the loss problem but they add noise (see Chapter 7) and worsen the impact of fiber dispersion and nonlinearity because signal degradation keeps on accumulating over multiple amplification stages. Indeed, periodically amplified lightwave systems are often limited by fiber dispersion unless dispersion-compensation techniques (discussed in Chapter 8) are used. Optoelectronic repeaters do not suffer from this problem as they regenerate the original bit stream and thus effectively compensate for all sources of signal degradation automatically. However, their repeated use in WDM systems (every 80 km or so) is not cost effective. Although considerable research effort is being directed toward developing all-optical regenerators (see Chapter 11), most terrestrial systems employ a combination of the two techniques shown in Figure 5.1 and place an optoelectronic regenerator after a certain number of optical amplifiers. Submarine systems are often designed to operate over a distance of more than 5000 km using only cascaded optical amplifiers.

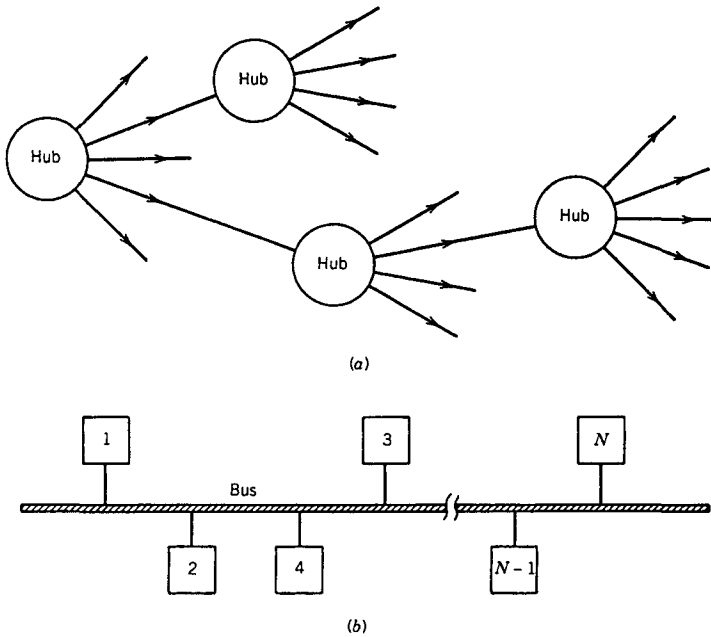


Figure 5.2: (a) Hub topology and (b) bus topology for distribution networks.

The spacing L between regenerators or optical amplifiers (see Figure 5.1), often called the *repeater spacing*, is a major design parameter simply because the system cost reduces as L increases. However, as discussed in Section 2.4, the distance L depends on the bit rate B because of fiber dispersion. The bit rate–distance product, BL , is generally used as a measure of the system performance for point-to-point links. The BL product depends on the operating wavelength, since both fiber losses and fiber dispersion are wavelength dependent. The first three generations of lightwave systems correspond to three different operating wavelengths near 0.85, 1.3, and 1.55 μm . Whereas the BL product was ~ 1 (Gb/s)-km for the first-generation systems operating near 0.85 μm , it becomes ~ 1 (Tb/s)-km for the third-generation systems operating near 1.55 μm and can exceed 1000 (Tb/s)-km for the fourth-generation systems.

5.1.2 Distribution Networks

Many applications of optical communication systems require that information is not only transmitted but also distributed to a group of subscribers. Examples include local-loop distribution of telephone services and broadcast of multiple video channels over cable television (CATV, short for common-antenna television). Considerable effort is directed toward the integration of audio and video services through a broadband digital network. The resulting bit stream can be transmitted using a variety of standards developed for this purpose. Transmission distances are relatively short ($L < 50$ km), but the bit rate can be as high as 100 Gb/s.

Figure 5.2 shows two topologies for distribution networks. In the case of *hub topology*, channel distribution takes place at central locations (or hubs), where an automated cross-connect facility switches channels in the electrical domain. Such networks are called *metropolitan-area networks* (MANs), or simply metro networks, as hubs are typically located in major cities [10]. The role of fiber is similar to the case of point-to-point links. Since the fiber bandwidth is generally much larger than that required by a single hub office, several offices can share a single fiber headed for the main hub. Telephone networks employ hub topology for distribution of audio channels within a city. A concern for the hub topology is related to its reliability—outage of a single fiber cable can affect the service to a large portion of the network. Additional point-to-point links can be used to guard against such a possibility by connecting important hub locations directly.

In the case of *bus topology*, a single fiber cable carries the multichannel optical signal throughout the area of service. Distribution is done by using optical taps, which divert a small fraction of the optical power to each subscriber. A simple CATV application of bus topology consists of distributing multiple video channels within a city. The use of optical fiber permits distribution of a large number of channels (100 or more) because of its large bandwidth compared with coaxial cables. The advent of *high-definition television* (HDTV) also requires lightwave transmission because of a large bandwidth associated with each video channel.

A problem with the bus topology is that the signal loss increases exponentially with the number of taps and limits the number of subscribers served by a single optical bus. Even when fiber losses are neglected, the power available at the N th tap is given by [1]

$$P_N = P_T C [(1 - \delta)(1 - C)]^{N-1}, \quad (5.1.1)$$

where P_T is the transmitted power, C is the fraction of power coupled out at each tap, and δ accounts for insertion losses, assumed to be the same at each tap. The derivation of Eq. (5.1.1) is left as an exercise for the reader. If we use $\delta = 0.05$, $C = 0.05$, $P_T = 1$ mW, and $P_N = 0.1$ μ W as illustrative values, N should not exceed 60. A solution to this problem is offered by optical amplifiers which can boost the optical power of the bus periodically and thus permit distribution to a large number of subscribers as long as the effects of fiber dispersion remain negligible.

5.1.3 Local-Area Networks

Many applications of fiber-optic communication technology require networks in which a large number of users within a local area (e.g., a university campus) are interconnected in such a way that any user can access the network randomly to transmit data to any other user [11]–[13]. Such networks are called *local-area networks* (LANs). Optical-access networks used in a local subscriber loop also fall in this category. Since the transmission distances are relatively short (<10 km), fiber losses are not of much concern for LAN applications. The major motivation behind the use of optical fibers is the large bandwidth offered by fiber-optic communication systems.

The main difference between MANs and LANs is related to the random access offered to multiple users of a LAN. The system architecture plays an important role

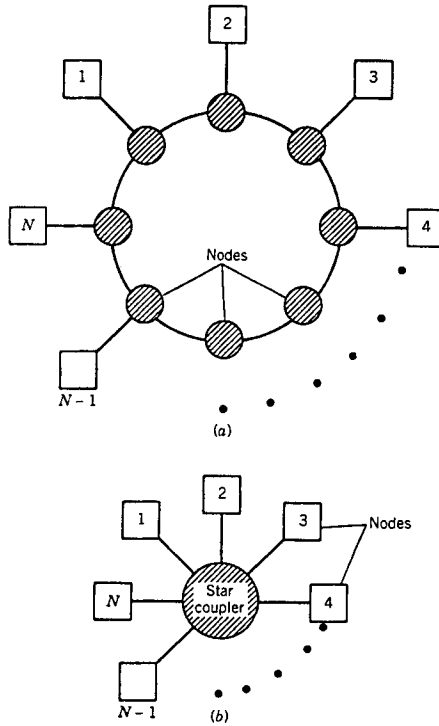


Figure 5.3: (a) Ring topology and (b) star topology for local-area networks.

for LANs, since the establishment of predefined protocol rules is a necessity in such an environment. Three commonly used topologies are known as bus, ring, and star configurations. The bus topology is similar to that shown in Figure 5.2(b). A well-known example of bus topology is provided by the *Ethernet*, a network protocol used to connect multiple computers and used by the *Internet*. The Ethernet operates at speeds of up to 10 Gb/s (10 GbE) by using a protocol based on *carrier-sense multiple access* (CSMA) with collision detection. A new standard known as 100 Gb Ethernet (officially IEEE 802.3ba) became operational in 2010. Its advent boosts the traffic speed on the Internet to a bit rate of 100 Gb/s. Figure 5.3 shows the ring and star topologies for LAN applications. In the ring topology [14], consecutive nodes are connected by point-to-point links to form a closed ring. Each node can transmit and receive the data by using a transmitter–receiver pair, which also acts as a repeater. A token (a predefined bit sequence) is passed around the ring. Each node monitors the bit stream to listen for its own address and to receive the data. It can also transmit by appending the data to an empty token. The use of ring topology for fiber-optic LANs has been commercialized with the standardized interface known as the fiber distributed data interface, FDDI for short [14].

In the *star topology* all nodes are connected through point-to-point links to a central node called a hub, or simply a star. Such LANs are further subclassified as *active-star*

or *passive-star* networks, depending on whether the central node is an active or passive device. In the active-star configuration, all incoming optical signals are converted to the electrical domain through optical receivers. The electrical signal is then distributed to drive individual node transmitters. Switching operations can also be performed at the central node since distribution takes place in the electrical domain. In the passive-star configuration, distribution takes place in the optical domain through devices such as directional couplers. Since the input from one node is distributed to many output nodes, the power transmitted to each node depends on the number of users. Similar to the case of bus topology, the number of users supported by passive-star LANs is limited by distribution losses. For an ideal $N \times N$ star coupler, the power reaching each node is simply P_T/N (if we neglect transmission losses) since the transmitted power P_T is divided equally among N users. For a passive star composed of directional couplers (see Section 8.2.4), the power is further reduced because of insertion losses and can be written as [1]

$$P_N = (P_T/N)(1 - \delta)^{\log_2 N}, \quad (5.1.2)$$

where δ is the insertion loss of each directional coupler. If we use $\delta = 0.05$, $P_T = 1$ mW, and $P_N = 0.1$ μ W as illustrative values, N can be as large as 500. This value of N should be compared with $N = 60$ obtained for the case of bus topology by using Eq. (5.1.1). A relatively large value of N makes star topology attractive for LAN applications.

5.2 Design Guidelines

The design of fiber-optic communication systems requires a clear understanding of the limitations imposed by the loss, dispersion, and nonlinearity of the fiber. Since fiber properties are wavelength dependent, the choice of operating wavelength is a major design issue. In this section we discuss how the bit rate and the transmission distance of a single-channel system are limited by fiber loss and dispersion; Chapter 6 is devoted to multichannel systems. We also consider the power and rise-time budgets and illustrate them through specific examples [9]. The power budget is also called the link budget, and the rise-time budget is sometimes referred to as the bandwidth budget.

5.2.1 Loss-Limited Lightwave Systems

Except for some short-haul fiber links, fiber losses play an important role in the system design. Consider an optical transmitter that is capable of launching an average power $\bar{P}_{tr}$. If the signal is detected by a receiver that requires a minimum average power $\bar{P}_{rec}$ at the bit rate B , the maximum transmission distance is limited by

$$L = \frac{10}{\alpha_f} \log_{10} \left(\frac{\bar{P}_{tr}}{\bar{P}_{rec}} \right), \quad (5.2.1)$$

where α_f is the net loss (in dB/km) of the fiber cable, including splice and connector losses. The bit-rate dependence of L arises from the linear dependence of $\bar{P}_{rec}$ on the bit rate B . Noting that $\bar{P}_{rec} = \bar{N}_p h \nu B$, where $h \nu$ is the photon energy and $\bar{N}_p$ is the average

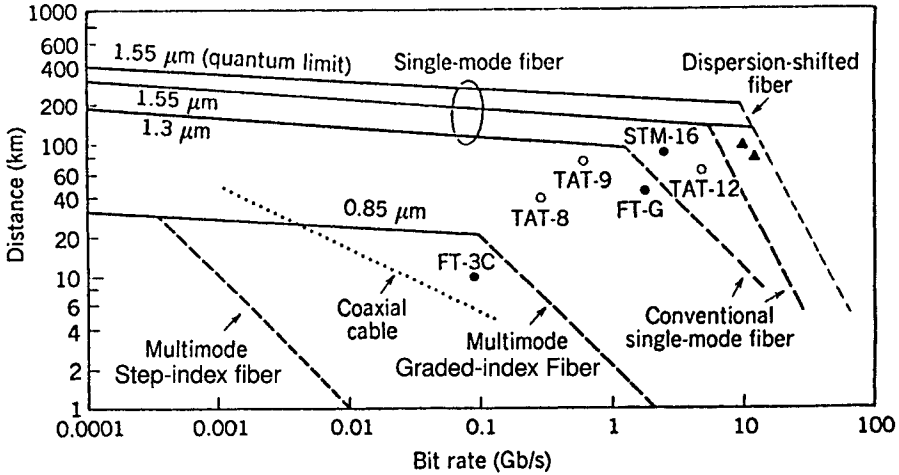


Figure 5.4: Loss (solid lines) and dispersion (dashed lines) limits on transmission distance L as a function of bit rate B for the three wavelength windows. The dotted line corresponds to coaxial cables. Circles denote commercial lightwave systems; triangles show laboratory experiments. (After Ref. [1]; ©1988 Academic Press; reprinted with permission.)

number of photons/bit required by the receiver, the distance L decreases logarithmically as B increases at a given operating wavelength.

The solid lines in Figure 5.4 show the dependence of L on B for three common operating wavelengths of 0.85, 1.3, and 1.55 μm by using $\alpha_f = 2.5, 0.4$, and 0.25 dB/km, respectively. The transmitted power is taken to be $\bar{P}_t = 1$ mW at the three wavelengths, whereas $\bar{N}_p = 300$ at $\lambda = 0.85$ μm and $\bar{N}_p = 500$ at 1.3 and 1.55 μm . The smallest value of L occurs for first-generation systems operating at 0.85 μm because of relatively large fiber losses near that wavelength. The repeater spacing of such systems is limited to 10–25 km, depending on the bit rate and the exact value of the loss parameter. In contrast, a repeater spacing of more than 100 km is possible for lightwave systems operating near 1.55 μm .

It is interesting to compare the loss limit of 0.85- μm lightwave systems with that of electrical communication systems based on coaxial cables. The dotted line in Figure 5.4 shows the bit-rate dependence of L for coaxial cables by assuming that the loss increases as $\sqrt{B}$. The transmission distance is larger for coaxial cables at small bit rates ($B < 5$ Mb/s), but fiber-optic systems take over at bit rates in excess of 5 Mb/s. Since a longer transmission distance translates into a smaller number of repeaters in a long-haul point-to-point link, fiber-optic communication systems offer an economic advantage when the operating bit rate exceeds 10 Mb/s.

The system requirements typically specified in advance are the bit rate B and the transmission distance L . The performance criterion is specified through the BER, a typical requirement being $\text{BER} < 10^{-9}$. The first decision of the system designer concerns the choice of the operating wavelength. As a practical matter, the cost of components is lowest near 0.85 μm and increases as wavelength shifts toward 1.3 and 1.55 μm .

Figure 5.4 can be quite helpful in determining the appropriate operating wavelength. Generally speaking, a fiber-optic link can operate near $0.85\text{ }\mu\text{m}$ if $B < 200\text{ Mb/s}$ and $L < 20\text{ km}$. This is the case for many LAN applications. On the other hand, the operating wavelength is by necessity in the $1.55\text{-}\mu\text{m}$ region for long-haul lightwave systems operating at bit rates in excess of 2 Gb/s . The curves shown in Figure 5.4 provide only a guide to the system design. Many other issues need to be addressed while designing a realistic fiber-optic communication system. Among them are the choice of the operating wavelength, selection of appropriate transmitters, receivers, and fibers, and the issues of cost, performance, and system reliability.

5.2.2 Dispersion-Limited Lightwave Systems

In Section 2.4 we discussed how fiber dispersion limits the bit rate–distance product BL because of pulse broadening. When the dispersion-limited transmission distance is shorter than the loss-limited distance of Eq. (5.2.1), the system is said to be dispersion-limited. The dashed lines in Figure 5.4 show the dispersion-limited transmission distance as a function of the bit rate. Since the physical mechanisms leading to dispersion limitation can be different for different operating wavelengths, let us examine each case separately.

Consider first the case of $0.85\text{-}\mu\text{m}$ lightwave systems, which often use multimode fibers to minimize the system cost. As discussed in Section 2.1, the most limiting factor for multimode fibers is intermodal dispersion. In the case of step-index multimode fibers, Eq. (2.1.6) provides an upper bound on the BL product. It is plotted in Figure 5.4 using $n_1 = 1.46$ and $\Delta = 0.01$. Even at a low bit rate of 1 Mb/s , such multimode systems are dispersion-limited, and their transmission distance is limited to below 10 km . For this reason, multimode step-index fibers are rarely used in the design of fiber-optic communication systems. Considerable improvement can be realized by using graded-index fibers for which intermodal dispersion limits the BL product to values given by Eq. (2.1.11). The condition $BL = 2c/(n_1\Delta^2)$ is plotted in Figure 5.4 and shows that $0.85\text{-}\mu\text{m}$ lightwave systems are loss-limited, rather than dispersion-limited, for bit rates up to 100 Mb/s when graded-index fibers are used. The first generation of terrestrial systems took advantage of such an improvement and used graded-index fibers. The first commercial system became available in 1980 and operated at a bit rate of 45 Mb/s with a repeater spacing of less than 10 km .

The second generation of lightwave systems used primarily single-mode fibers near the minimum-dispersion wavelength occurring at about $1.31\text{ }\mu\text{m}$. The most limiting factor for such systems is dispersion-induced pulse broadening dominated by a relatively large source spectral width. As discussed in Section 2.4.3, the BL product is then limited by Eq. (2.4.26). The value of $|D|$ depends on how close the operating wavelength is to the zero-dispersion wavelength of the fiber and is typically $\sim 1\text{ ps/(km}\cdot\text{nm)}$. Figure 5.4 shows the dispersion limit for $1.3\text{-}\mu\text{m}$ lightwave systems by choosing $|D|\sigma_\lambda = 2\text{ ps/km}$ so that $BL \leq 125\text{ (Gb/s)}\cdot\text{km}$. As seen there, such systems are generally loss-limited for bit rates up to 1 Gb/s but become dispersion-limited at higher bit rates.

Third- and fourth-generation lightwave systems operate near $1.55\text{ }\mu\text{m}$ to take advantage of the smallest fiber losses occurring in this wavelength region. However,

dispersion becomes a major problem for such systems because $D \approx 16$ ps/(km-nm) near $1.55 \mu\text{m}$ for standard silica fibers. Semiconductor lasers operating in a single longitudinal mode provide a solution to this problem. The dispersion limit is then given by Eq. (2.4.30). Figure 5.4 shows this limit by choosing $B^2L = 4000$ (Gb/s)²-km. As seen there, such $1.55\text{-}\mu\text{m}$ systems become dispersion-limited only for $B > 5$ Gb/s. In practice, the frequency chirp imposed on the optical pulse during direct modulation provides a much more severe limitation. Qualitatively speaking, the frequency chirp manifests through a broadening of the pulse spectrum. If we use Eq. (2.4.26) with $D = 16$ ps/(km-nm) and $\sigma_\lambda = 0.1$ nm, the BL product is limited to 150 (Gb/s)-km. As a result, the frequency chirp limits the transmission distance to 75 km at $B = 2$ Gb/s, even though loss-limited distance exceeds 150 km. The frequency-chirp problem is often solved by using an external modulator for systems operating at bit rates > 5 Gb/s.

A solution to the dispersion problem is offered by *dispersion-shifted fibers* for which dispersion and loss both are minimum near $1.55 \mu\text{m}$. Figure 5.4 shows the improvement by using Eq. (2.4.30) with $|\beta_2| = 2$ ps²/km. Such systems can be operated at 20 Gb/s with a repeater spacing of about 80 km. Further improvement is possible by operating the lightwave system very close to the zero-dispersion wavelength, a task that is not always feasible because of variations in the dispersive properties of the fiber along the transmission link. In practice, the frequency chirp makes it difficult to achieve even the limit indicated in Figure 5.4. By 1989, two laboratory experiments had demonstrated transmission over 81 km at 11 Gb/s [15] and over 100 km at 10 Gb/s [16] by using low-chirp semiconductor lasers together with dispersion-shifted fibers. The triangles in Figure 5.4 show that such systems operate quite close to the fundamental limits set by fiber dispersion. Transmission over longer distances requires the use of dispersion management, a technique discussed in Chapter 8.

5.2.3 Power Budget

The purpose of the *power budget* is to ensure that enough power will reach the receiver to maintain reliable performance during the entire system lifetime. The minimum average power required by the receiver is the receiver sensitivity $\bar{P}_{\text{rec}}$ (see Section 4.6). The average launch power $\bar{P}_{\text{tr}}$ is generally known for any transmitter. The power budget takes an especially simple form in decibel units with optical powers expressed in dBm units (see Appendix A). More specifically,

$$\bar{P}_{\text{tr}} = \bar{P}_{\text{rec}} + C_L + M_s, \quad (5.2.2)$$

where C_L is the total channel loss and M_s is the *system margin*. The purpose of system margin is to allocate a certain amount of power to additional sources of power penalty that may develop during system lifetime because of component degradation or other unforeseen events. A system margin of 3–4 dB is typically allocated during the design process.

The channel loss C_L should take into account all possible sources of power loss, including connector and splice losses. If α_f is the fiber loss in decibels per kilometer, C_L can be written as

$$C_L = \alpha_f L + \alpha_{\text{con}} + \alpha_{\text{splice}}, \quad (5.2.3)$$

Table 5.1 Power budget of a 0.85- μm lightwave system

Quantity	Symbol	Laser	LED
Transmitter power	$\bar{P}_{\text{tr}}$	0 dBm	-13 dBm
Receiver sensitivity	$\bar{P}_{\text{rec}}$	-42 dBm	-42 dBm
System margin	M_s	6 dB	6 dB
Available channel loss	C_L	36 dB	23 dB
Connector loss	α_{con}	2 dB	2 dB
Fiber cable loss	α_f	3.5 dB/km	3.5 dB/km
Maximum fiber length	L	9.7 km	6 km

where α_{con} and α_{splice} account for the connector and splice losses throughout the fiber link. Sometimes splice loss is included within the specified loss of the fiber cable. The connector loss α_{con} includes connectors at the transmitter and receiver ends but must include other connectors if used within the fiber link.

Equations (5.2.2) and (5.2.3) can be used to estimate the maximum transmission distance for a given choice of the components. As an illustration, consider the design of a fiber link operating at 100 Mb/s and requiring a maximum transmission distance of 8 km. As seen in Figure 5.4, such a system can be designed to operate near 0.85 μm provided that a graded-index multimode fiber is used for the optical cable. The operation near 0.85 μm is desirable from the economic standpoint. Once the operating wavelength is selected, a decision must be made about the appropriate transmitters and receivers. The GaAs transmitter can use a semiconductor laser or an LED as an optical source. Similarly, the receiver can be designed to use either a $p-i-n$ or an avalanche photodiode. Keeping the low cost in mind, let us choose a $p-i-n$ receiver and assume that it requires 2500 photons/bit on average to operate reliably with a BER below 10^{-9} . Using the relation $\bar{P}_{\text{rec}} = \bar{N}_p h \nu B$ with $\bar{N}_p = 2500$ and $B = 100$ Mb/s, the receiver sensitivity is given by $\bar{P}_{\text{rec}} = -42$ dBm. The average launch power for LED and laser-based transmitters is typically 50 μW and 1 mW, respectively.

Table 5.1 shows the power budget for the two transmitters by assuming that the splice loss is included within the cable loss. The transmission distance L is limited to 6 km in the case of LED-based transmitters. If the system specification is 8 km, a more expensive laser-based transmitter must be used. The alternative is to use an avalanche photodiode (APD) receiver. If the receiver sensitivity improves by more than 7 dB when an APD is used in place of a $p-i-n$ photodiode, the transmission distance can be increased to 8 km even for an LED-based transmitter. Economic considerations would then dictate the choice between the laser-based transmitters and APD receivers.

5.2.4 Rise-Time Budget

The purpose of the *rise-time budget* is to ensure that the system is able to operate properly at the intended bit rate. Even if the bandwidth of the individual system components exceeds the bit rate, it is still possible that the total system may not be able to operate at that bit rate. The concept of rise time is used to allocate the bandwidth among various components. The rise time T_r of a linear system is defined as the time during which the

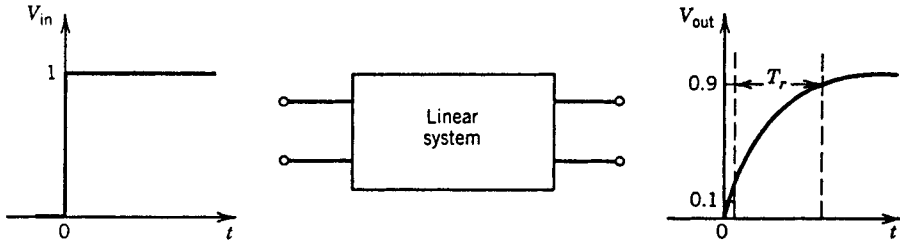


Figure 5.5: Rise time T_r associated with a bandwidth-limited linear system.

response increases from 10 to 90% of its final output value when the input is changed abruptly. Figure 5.5 illustrates the concept graphically.

An inverse relationship exists between the bandwidth Δf and the rise time T_r associated with a linear system. This relationship can be understood by considering a simple RC circuit as an example of the linear system. When the input voltage across an RC circuit changes instantaneously from 0 to V_0 , the output voltage changes as

$$V_{\text{out}}(t) = V_0[1 - \exp(-t/RC)], \quad (5.2.4)$$

where R is the resistance and C is the capacitance of the RC circuit. The rise time is found to be given by

$$T_r = (\ln 9)RC \approx 2.2RC. \quad (5.2.5)$$

The transfer function $H(f)$ of the RC circuit is obtained by taking the Fourier transform of Eq. (5.2.4) and is of the form

$$H(f) = (1 + i2\pi fRC)^{-1}. \quad (5.2.6)$$

The bandwidth Δf of the RC circuit corresponds to the frequency at which $|H(f)|^2 = 1/2$ and is given by the well-known expression $\Delta f = (2\pi RC)^{-1}$. By using Eq. (5.2.5), Δf and T_r are related as

$$T_r = \frac{2.2}{2\pi\Delta f} = \frac{0.35}{\Delta f}. \quad (5.2.7)$$

The inverse relationship between the rise time and the bandwidth is expected to hold for any linear system. However, the product $T_r\Delta f$ would generally be different than 0.35. One can use $T_r\Delta f = 0.35$ in the design of optical communication systems as a conservative guideline. The relationship between the bandwidth Δf and the bit rate B depends on the digital format. In the case of return-to-zero (RZ) format (see Section 1.2), $\Delta f = B$ and $BT_r = 0.35$. By contrast, $\Delta f \approx B/2$ for the nonreturn-to-zero (NRZ) format and $BT_r = 0.7$. In both cases, the specified bit rate imposes an upper limit on the maximum rise time that can be tolerated. The communication system must be designed to ensure that T_r is below this maximum value, i.e.,

$$T_r \leq \begin{cases} 0.35/B & \text{for RZ format,} \\ 0.70/B & \text{for NRZ format.} \end{cases} \quad (5.2.8)$$

The three components of fiber-optic communication systems have individual rise times. The total rise time of the whole system is related to the individual component rise times approximately as [17]

$$T_r^2 = T_{tr}^2 + T_{fiber}^2 + T_{rec}^2, \quad (5.2.9)$$

where T_{tr} , T_{fiber} , and T_{rec} are the rise times associated with the transmitter, fiber, and receiver, respectively. The rise times of the transmitter and the receiver are generally known to the system designer. The transmitter rise time T_{tr} is determined primarily by the electronic components of the driving circuit and the electrical parasitics associated with the optical source. Typically, T_{tr} is a few nanoseconds for LED-based transmitters but can be shorter than 0.1 ns for laser-based transmitters. The receiver rise time T_{rec} is determined primarily by the 3-dB electrical bandwidth of the receiver front end. Equation (5.2.7) can be used to estimate T_{rec} if the front-end bandwidth is specified.

The fiber rise time T_{fiber} should in general include the contributions of both the intermodal dispersion and group-velocity dispersion (GVD) through the relation

$$T_{fiber}^2 = T_{modal}^2 + T_{GVD}^2. \quad (5.2.10)$$

For single-mode fibers, $T_{modal} = 0$ and $T_{fiber} = T_{GVD}$. In principle, one can use the concept of fiber bandwidth discussed in Section 2.4.4 and relate T_{fiber} to the 3-dB fiber bandwidth f_{3dB} through a relation similar to Eq. (5.2.7). In practice it is not easy to calculate f_{3dB} , especially in the case of modal dispersion. The reason is that a fiber link consists of many concatenated fiber sections (typical length 5 km), which may have different dispersion characteristics. Furthermore, mode mixing occurring at splices and connectors tends to average out the propagation delay associated with different modes of a multimode fiber. A statistical approach is often necessary to estimate the fiber bandwidth and the corresponding rise time [18]–[21].

In a phenomenological approach, T_{modal} can be approximated by the time delay ΔT given by Eq. (2.1.5) in the absence of mode mixing, i.e.,

$$T_{modal} \approx (n_1 \Delta / c) L, \quad (5.2.11)$$

where $n_1 \approx n_2$ was used. For graded-index fibers, Eq. (2.1.10) is used in place of Eq. (2.1.5), resulting in $T_{modal} \approx (n_1 \Delta^2 / 8c) L$. In both cases, the effect of mode mixing is included by changing the linear dependence on L by a sublinear dependence L^q , where q has a value in the range 0.5–1, depending on the extent of mode mixing. A reasonable estimate based on the experimental data is $q = 0.7$. The contribution T_{GVD} can also be approximated by ΔT given by Eq. (2.3.4), so that

$$T_{GVD} \approx |D| L \Delta \lambda, \quad (5.2.12)$$

where $\Delta \lambda$ is the spectral width of the optical source (taken as a full width at half maximum). The dispersion parameter D may change along the fiber link if different sections have different dispersion characteristics; an average value should be used in Eq. (5.2.12) in that case.

As an illustration of the rise-time budget, consider a 1.3- μm lightwave system designed to operate at 1 Gb/s over a single-mode fiber with a repeater spacing of 50 km.

The rise times for the transmitter and the receiver have been specified as $T_{tr} = 0.25$ ns and $T_{rec} = 0.35$ ns. The source spectral width is specified as $\Delta\lambda = 3$ nm, whereas the average value of D is 2 ps/(km-nm) at the operating wavelength. From Eq. (5.2.12), $T_{GVD} = 0.3$ ns for a link length $L = 50$ km. Modal dispersion does not occur in single-mode fibers. Hence $T_{modal} = 0$ and $T_{fiber} = 0.3$ ns. The system rise time is estimated by using Eq. (5.2.9) and is found to be $T_r = 0.524$ ns. The use of Eq. (5.2.8) indicates that such a system cannot be operated at 1 Gb/s when the RZ format is employed for the optical bit stream. However, it would operate properly if digital format is changed to the NRZ format. If the use of RZ format is a prerequisite, the designer must choose different transmitters and receivers to meet the rise-time budget requirement. The use of the NRZ format was prevalent during the 1990s, but the RZ format has become the format of choice for long-haul systems operating at a bit rate of 40 Gb/s, especially when the information is coded in the phase of the optical carrier.

5.3 Long-Haul Systems

With the advent of optical amplifiers, fiber losses can be compensated by inserting amplifiers periodically along a long-haul fiber link (see Figure 5.1). At the same time, the effects of fiber dispersion (GVD) can be reduced by using dispersion management (see Chapter 8). Since neither the fiber loss nor the GVD is then a limiting factor, one may ask how many in-line amplifiers can be cascaded in series, and what limits the total link length. This topic is covered in Chapter 7. Here we focus on the factors that limit the performance of amplified fiber links and provide a few design guidelines. The section also outlines the progress realized in the development of terrestrial and undersea lightwave systems since 1980 when the first system was installed.

5.3.1 Performance-Limiting Factors

The most important consideration in designing a periodically amplified fiber link is related to the *nonlinear effects* occurring inside all optical fibers [22] (see Section 2.6). For single-channel lightwave systems, the dominant nonlinear phenomenon that limits the system performance is *self-phase modulation* (SPM). When optoelectronic regenerators are used, the SPM effects accumulate only over one repeater spacing (typically <100 km) and are of little concern if the launch power satisfies Eq. (2.6.15) with $N_A = 1$, or the condition $P_{in} \ll 22$ mW. In contrast, the SPM effects accumulate over long lengths (~ 1000 km) when in-line amplifiers are used periodically for loss compensation. A rough estimate of the limitation imposed by the SPM is again obtained from Eq. (2.6.15). This equation predicts that the peak power should be below 2.2 mW for 10 cascaded amplifiers when the nonlinear parameter $\gamma = 2$ W⁻¹/km. The condition on the average power depends on the modulation format and the shape of optical pulses. It is nonetheless clear that the average power should be reduced to below 1 mW for SPM effects to remain negligible for a lightwave system designed to operate over a distance of more than 1000 km. The limiting value of the average power also depends on the type of fiber in which light is propagating through the effective core area A_{eff} .

The SPM effects are most dominant inside dispersion-compensating fibers for which A_{eff} is typically close to $20 \mu\text{m}^2$.

The forgoing discussion of the SPM-induced limitations is too simplistic to be accurate since it completely ignores the role of fiber dispersion. In fact, as the dispersive and nonlinear effects act on the optical signal simultaneously, their mutual interplay becomes quite important [22]. The effect of SPM on pulses propagating inside an optical fiber can be included by using the nonlinear Schrödinger (NLS) equation of Section 2.6. This equation is given by [see Eq. (2.6.18)]

$$\frac{\partial A}{\partial z} + \frac{i\beta_2}{2} \frac{\partial^2 A}{\partial t^2} = -\frac{\alpha}{2} A + i\gamma |A|^2 A, \quad (5.3.1)$$

where fiber losses are included through the α term. This term can also include periodic amplification of the signal by treating α as a function of z . The NLS equation is used routinely for designing modern lightwave systems.

Because of the nonlinear nature of Eq. (5.3.1), it should be solved numerically in general. A numerical approach is indeed adopted (see Appendix D) for quantifying the impact of SPM on the performance of long-haul lightwave systems [23]–[31]. The use of a large-effective-area fiber (LEAF) helps by reducing the nonlinear parameter γ defined as $\gamma = 2\pi n_2 / (\lambda A_{\text{eff}})$. Appropriate chirping of input pulses can also be beneficial for reducing the SPM effects. This feature has led to the adoption of a new modulation format known as the chirped RZ or CRZ format. Numerical simulations show that, in general, the launch power must be optimized to a value that depends on many design parameters such as the bit rate, total link length, and amplifier spacing. In one study, the optimum launch power was found to be about 1 mW for a 5-Gb/s signal transmitted over 9000 km with 40-km amplifier spacing [27].

The combined effects of GVD and SPM also depend on the sign of the dispersion parameter β_2 . In the case of anomalous dispersion ($\beta_2 < 0$), the nonlinear phenomenon of *modulation instability* [22] can affect the system performance drastically [28]. This problem can be overcome by using a combination of fibers with normal and anomalous GVD such that the average dispersion over the entire fiber link is “normal.” However, a new kind of modulation instability, referred to as *sideband instability* [32], can occur in both the normal and anomalous GVD regions. It has its origin in the periodic variation of the signal power along the fiber link when equally spaced optical amplifiers are used to compensate for fiber losses. Since the quantity $\gamma |A|^2$ in Eq. (5.3.1) is then a periodic function of z , the resulting nonlinear-index grating can initiate a four-wave-mixing process that generates sidebands in the signal spectrum. It can be avoided by making the amplifier spacing nonuniform.

Another factor that plays a crucial role is the noise added by optical amplifiers. Similar to the case of electronic amplifiers, the noise of optical amplifiers is quantified through an amplifier noise figure F_n (see Chapter 7). The nonlinear interaction between the amplified spontaneous emission and the signal can lead to a large spectral broadening through the nonlinear phenomena such as cross-phase modulation and four-wave mixing [33]. Because the noise has a much larger bandwidth than the signal, its impact can be reduced by using optical filters. Numerical simulations indeed show a considerable improvement when optical filters are used after every in-line amplifier [27].

The polarization effects that are totally negligible in the traditional “nonamplified” lightwave systems become of concern for long-haul systems with in-line amplifiers. The polarization-mode dispersion (PMD) issue has been discussed in Section 2.3.5. In addition to PMD, optical amplifiers can also induce polarization-dependent gain and loss [26]. Although the PMD effects must be considered during system design, their impact depends on the design parameters such as the bit rate and the transmission distance. For bit rates as high as 10-Gb/s, the PMD effects can be reduced to an acceptable level with a proper design. However, PMD becomes of major concern for 40-Gb/s systems for which the bit slot is only 25 ps wide. The use of a PMD-compensation technique is often necessary at such high bit rates.

The fourth generation of lightwave systems began in 1995 when lightwave systems employing amplifiers first became available commercially. Of course, the laboratory demonstrations began as early as 1989. Many experiments used a recirculating fiber loop to demonstrate system feasibility as it was not practical to use long lengths of fiber in a laboratory setting. Already in 1991, an experiment showed the possibility of data transmission over 21,000 km at 2.5 Gb/s, and over 14,300 km at 5 Gb/s, by using the recirculating-loop configuration [34]. In a system trial carried out in 1995 by using actual submarine cables and repeaters [35], a 5.3-Gb/s signal was transmitted over 11,300 km with 60 km of amplifier spacing. This system trial led to the deployment of a commercial transpacific cable (TPC-5) that began operating in 1996.

The bit rate of fourth-generation systems was extended to 10 Gb/s beginning in 1992. As early as 1995, a 10-Gb/s signal was transmitted over 6480 km with 90-km amplifier spacing [36]. With a further increase in the distance, the SNR decreased below the value needed to maintain the BER below 10^{-9} . One may think that the performance should improve by operating close to the zero-dispersion wavelength of the fiber. However, an experiment, performed under such conditions, achieved a distance of only 6000 km at 10 Gb/s even with 40-km amplifier spacing [37], and the situation became worse when the RZ modulation format was used. Starting in 1999, the single-channel bit rate was pushed toward 40 Gb/s in several experiments [38]–[40], and by 2002 such systems became available commercially. The design of 40-Gb/s lightwave systems requires the use of several new ideas including the CRZ format, dispersion management with GVD-slope compensation, and distributed Raman amplification. Even then, the combined effects of the higher-order dispersion, PMD, and SPM degrade the system performance considerably at a bit rate of 40 Gb/s.

5.3.2 Terrestrial Lightwave Systems

An important application of fiber-optic communication links is for enhancing the capacity of telecommunication networks worldwide. Indeed, it is this application that started the field of optical fiber communications in 1977 and has propelled it since then by demanding systems with higher and higher capacities. Here we focus on the status of commercial systems by considering the terrestrial and undersea systems separately.

After a successful Chicago field trial in 1977, terrestrial lightwave systems became available commercially beginning in 1980 [41]–[43]. Table 5.2 lists the operating characteristics of several terrestrial systems developed since then. The first-generation systems operated near $0.85 \mu\text{m}$ and used multimode graded-index fibers as the trans-

Table 5.2 Terrestrial lightwave systems

System	Year	λ (μm)	B (Mb/s)	L (km)	Voice Channels
FT-3	1980	0.85	45	< 10	672
FT-3C	1983	0.85	90	< 15	1,344
FT-3X	1984	1.30	180	< 25	2,688
FT-G	1985	1.30	417	< 40	6,048
FT-G-1.7	1987	1.30	1,668	< 46	24,192
STM-16	1991	1.55	2,488	< 85	32,256
STM-64	1996	1.55	9,953	< 90	129,024
STM-256	2002	1.55	39,813	< 90	516,096

mission medium. As seen in Figure 5.4, the BL product of such systems is limited to 2 (Gb/s)-km. A commercial lightwave system (FT-3C) operating at 90 Mb/s with a repeater spacing of about 12 km realized a BL product of nearly 1 (Gb/s)-km; it is shown by a filled circle in Figure 5.4. The operating wavelength moved to 1.3 μm in second-generation lightwave systems to take advantage of low fiber losses and low dispersion near this wavelength. The BL product of 1.3- μm lightwave systems is limited to about 100 (Gb/s)-km when a multimode semiconductor laser is used inside the transmitter. In 1987, a commercial 1.3- μm lightwave system provided data transmission at 1.7 Gb/s with a repeater spacing of about 45 km. A filled circle in Figure 5.4 shows that this system operates quite close to the dispersion limit.

The third generation of lightwave systems became available commercially in 1991. Such systems operate near 1.55 μm at bit rates in excess of 2 Gb/s, typically at 2.488 Gb/s, corresponding to the OC-48 level of the SONET, or the STS-16 level of the SDH, specifications. The switch to the 1.55- μm wavelength helps to increase the loss-limited transmission distance to more than 100 km because of fiber losses of less than 0.25 dB/km in this wavelength region. However, the repeater spacing was limited to below 100 km because of the high GVD of standard telecommunication fibers. In fact, the deployment of third-generation lightwave systems was possible only after the development of distributed feedback (DFB) semiconductor lasers, which reduce the impact of fiber dispersion by reducing the source spectral width to below 100 MHz (see Section 2.4).

The fourth generation of lightwave systems appeared around 1996. Such systems operate in the 1.55- μm region at a bit rate as high as 40 Gb/s by using dispersion-shifted fibers in combination with optical amplifiers. However, more than 50 million kilometers of the standard telecommunication fiber is already installed in the worldwide telephone network. Economic reasons dictate that the fourth generation of lightwave systems make use of this existing base. Two approaches are being used to solve the dispersion problem. First, several dispersion-management schemes (discussed in Chapter 8) make it possible to extend the bit rate to 10 Gb/s while maintaining an amplifier spacing of up to 100 km. Second, several 10-Gb/s signals can be transmitted simultaneously by using the WDM technique discussed in Chapter 6. Moreover, if the WDM technique is combined with dispersion management, the total transmission

distance can approach several thousand kilometers provided that fiber losses are compensated periodically by using optical amplifiers. Such WDM lightwave systems were deployed commercially worldwide beginning in 1996 and allowed a system capacity of 1.6 Tb/s by 2000 for the 160-channel commercial WDM systems.

The fifth generation of lightwave systems began to emerge around 2001 [44]–[52]. The bit rate of each channel in this generation of WDM systems is 40 Gb/s (corresponding to the STM-256 or OC-768 level). Several new techniques developed in recent years make it possible to transmit a 40-Gb/s optical signal over long distances. New dispersion-shifted fibers have been developed with smaller PMD levels. Their use in combination with tunable dispersion-compensating techniques can compensate the GVD for all channels simultaneously. The use of Raman amplification helps to reduce the noise and improves the SNR at the receiver. The use of a forward-error correction (FEC) technique (see Section 5.5) helps to increase the transmission distance by reducing the required SNR. The number of WDM channels can be increased by using the L and S bands located on the long- and short-wavelength sides of the conventional C band occupying the 1530–1570-nm spectral region. In a 2001 experiment, 77 channels, each operating at 42.7-Gb/s, were transmitted over 1200 km by using the C and L bands simultaneously, resulting in a 3-Tb/s capacity [44]. In another 2001 experiment, the system capacity was extended to 10.2 Tb/s by transmitting 256 channels over 100 km at 42.7 Gb/s per channel using only the C and L bands, resulting in a spectral efficiency of 1.28 (b/s)/Hz [45]. The bit rate was 42.7 Gb/s in both of these experiments because of the overhead associated with the FEC technique.

Starting in 2002, the research focus shifted toward advanced modulation formats in which information is coded using optical phase rather than amplitude of the carrier wave (see Chapter 10). This approach has led to considerable improvement in the spectral efficiency of WDM systems. In a 2007 experiment [52], 25.6-Tb/s transmission was realized over 240 km of optical fibers using 160 WDM channels that spanned both the C and L bands with 50-GHz channel spacing. Each channel contained two polarization-multiplexed 85.4-Gb/s signals coded with the DQPSK format, resulting in a spectral efficiency of 3.2 b/s/Hz. By 2010, transmission at a total bit rate of 69.1 Tb/s was demonstrated over 240 km of fiber using 432 WDM channels, each operating at 171 Gb/s with a 7% FEC overhead [53].

5.3.3 Undersea Lightwave Systems

Undersea or submarine transmission systems are used for intercontinental communications and are capable of providing a network spanning the whole earth [54]–[56]. Reliability is of major concern for such systems as repairs are expensive. Generally, undersea systems are designed for a 25-year service life, with at most three failures during operation. Figure 1.5 shows the multitude of undersea systems deployed worldwide. Table 5.3 lists several high-capacity undersea fiber-optic cable systems installed after the year 2000. Most of them transport multiple WDM channels, each operating at 10 Gb/s, and employ several fiber pairs within each cable to further enhance the system capacity to beyond 1 Tb/s.

The first undersea fiber-optic cable (TAT-8) was a second-generation system. It was installed in 1988 in the Atlantic Ocean, with a repeater spacing of up to 70 km,

Table 5.3 High-capacity undersea lightwave systems

System Name	Year	Capacity (Gb/s)	Length (km)	WDM Channels	Fiber Pairs
TAT-14	2001	640	15,428	16	4
SEA-ME-WE 3	2001	960	39,000	48	2
AC-2	2001	1280	6,400	32	4
VSNL Transatlantic	2001	2560	13,000	64	4
FLAG	2001	4800	28,000	60	8
Apollo	2003	3200	13,000	80	4
SEA-ME-WE 4	2005	1280	18,800	64	2
Asia-America Gateway	2008	1920	20,000	96	2
India-ME-WE	2009	3840	13,000	96	4

and transported a single channel a bit rate of 280 Mb/s. The system design was on the conservative side, mainly to ensure reliability. The same technology was used for the first transpacific lightwave system (TPC-3), which became operational in 1989. By 1990 the third-generation lightwave systems had been developed. The TAT-9 submarine system used this technology in 1991; it was designed to operate near $1.55 \mu\text{m}$ at a bit rate of 560 Mb/s with a repeater spacing of about 80 km. The increasing traffic across the Atlantic Ocean led to the deployment of the TAT-10 and TAT-11 lightwave systems by 1993 with the same technology.

The advent of optical amplifiers prompted their use in the next generation of undersea systems. The TAT-12 cable, installed in 1995, employed optical amplifiers in place of optoelectronic regenerators and operated at a bit rate of 5.3 Gb/s with an amplifier spacing of about 50 km. The bit rate was slightly larger than the STM-32 bit rate of 5 Gb/s because of the overhead associated with the forward-error-correction technique discussed in Section 5.5. The design of such lightwave systems becomes quite complex because of the cumulative effects of fiber dispersion and nonlinearity, which must be controlled over long distances. The transmitter power and the dispersion profile along the link must be optimized to combat such effects.

A second category of undersea lightwave systems requires repeaterless transmission over several hundred kilometers [55]. Such systems are used for interisland communication or for looping a shoreline such that the signal is regenerated on the shore periodically after a few hundred kilometers of undersea transmission. The dispersive and nonlinear effects are of less concern for such systems than for transoceanic lightwave systems, but fiber losses become a major issue. The reason is easily appreciated by noting that the cable loss exceeds 100 dB over a distance of 500 km even under the best operating conditions. In the 1990s several laboratory experiments demonstrated repeaterless transmission at 2.5 Gb/s over more than 500 km by using two in-line amplifiers that were pumped remotely from the transmitter and receiver ends with high-power pump lasers. Another amplifier at the transmitter boosted the launched power to close to 100 mW.

Such high input powers exceed the threshold level for stimulated Brillouin scattering (SBS), a nonlinear phenomenon discussed in Section 2.6. The suppression of SBS

is often realized by modulating the phase of the optical carrier such that the carrier linewidth is broadened to 200 MHz or more from its initial value of <10 MHz [57]. Directly modulated DFB lasers can also be used for this purpose. In a 1996 experiment, a 2.5-Gb/s signal was transmitted over 465 km by direct modulation of a DFB laser [58]. Chirping of the modulated signal broadened the spectrum enough that an external phase modulator was not required provided that the launched power was kept below 100 mW. The bit rate of repeaterless undersea systems can be increased to 10 Gb/s by employing the same techniques used at 2.5 Gb/s. In a 1996 experiment [59], the 10-Gb/s signal was transmitted over 442 km by using two remotely pumped in-line amplifiers. Two external modulators were used, one for SBS suppression and another for signal generation. In a 1998 experiment, a 40-Gb/s signal was transmitted over 240 km using the RZ format and an alternating polarization format [60].

The use of the WDM technique in combination with optical amplifiers, dispersion management, and error correction has revolutionized the design of submarine fiber-optic systems [61]–[68]. In 1998, a submarine cable known as AC-1 was deployed across the Atlantic Ocean with a capacity of 80 Gb/s using the WDM technology. An identically designed system (PC-1) crossed the Pacific Ocean. The use of dense WDM, in combination with multiple fiber pairs per cable, resulted in systems with large capacities. By 2001, several systems with a capacity of >1 Tb/s became operational across the Atlantic Ocean (see Table 5.3). These systems employ a ring configuration and cross the Atlantic Ocean twice to ensure fault tolerance. The VSNL Transatlantic submarine system can achieve a total capacity of 2.56 Tb/s and spans a total distance of 13,000 km. Another system, known as Apollo, is capable of carrying traffic at speeds of up to 3.2 Tb/s by transmitting 80 channels (each operating at 10 Gb/s) over 4 fiber pairs.

The pace slowed down after 2001 with the bursting of the “telecom bubble.” However, the development of undersea systems has continued within industrial laboratories. In a 2003 experiment, transmission over 9400 km of 40 channels (each operating at 42.7 Gb/s with a 70-GHz channel spacing) was realized using phase modulation (with the DPSK format), FEC coding, and distributed Raman amplification [62]. By 2009, another experiment transmitted 72 channels, each operating at 100 Gb/s, over 10,000 km using the QPSK modulation format with digital processing in a coherent receiver [63]. On the commercial side, a field trial was carried out as early as 2004 in which 96 channels at 10 Gb/s were transmitted successfully over a distance of 13,000 km [64]. As seen in Table 5.3, several new transoceanic systems have been deployed worldwide in recent years. A few others such as Europe–India Gateway were in various stages of completion in 2010.

5.4 Sources of Power Penalty

The sensitivity of the optical receiver in a realistic lightwave system is affected by several physical phenomena which, in combination with fiber dispersion, degrade the SNR at the decision circuit. Among the phenomena that degrade the receiver sensitivity are modal noise, dispersion broadening and intersymbol interference, mode-partition noise, frequency chirp, and reflection feedback. In this section we discuss how the

system performance is affected by fiber dispersion by considering the extent of power penalty resulting from these phenomena.

5.4.1 Modal Noise

Modal noise is associated with multimode fibers and was studied extensively during the 1980s [69]–[82]. Its origin can be understood as follows. Interference among various propagating modes in a multimode fiber creates a *speckle pattern* at the photodetector. The nonuniform intensity distribution associated with the speckle pattern is harmless in itself, as the receiver performance is governed by the total power integrated over the detector area. However, if the speckle pattern fluctuates with time, it will lead to fluctuations in the received power that would degrade the SNR. Such fluctuations are referred to as *modal noise*. They invariably occur in multimode fiber links because of mechanical disturbances such as vibrations and microbends. In addition, splices and connectors act as *spatial filters*. Any temporal changes in spatial filtering translate into speckle fluctuations and enhancement of the modal noise. Modal noise is strongly affected by the source spectral bandwidth $\Delta\nu$ since mode interference occurs only if the coherence time ($T_c \approx 1/\Delta\nu$) is longer than the intermodal delay time ΔT given by Eq. (2.1.5). For LED-based transmitters $\Delta\nu$ is large enough ($\Delta\nu \sim 5$ THz) that this condition is not satisfied. Most lightwave systems that use multimode fibers also use LEDs to avoid the modal-noise problem.

Modal noise becomes a serious problem when semiconductor lasers are used in combination with multimode fibers. Attempts have been made to estimate the extent of sensitivity degradation induced by modal noise [71]–[73] by calculating the BER after adding modal noise to the other sources of receiver noise. Figure 5.6 shows the power penalty at a BER of 10^{-12} calculated for a 1.3- μm lightwave system operating at 140 Mb/s. The graded-index fiber has a 50- μm core diameter and supports 146 modes. The power penalty depends on the mode-selective coupling loss occurring at splices and connectors. It also depends on the longitudinal-mode spectrum of the semiconductor laser. As expected, power penalty decreases as the number of longitudinal modes increases because of a reduction in the coherence time of the emitted light.

Modal noise can also occur in single-mode systems if short sections of fiber are installed between two connectors or splices during repair or normal maintenance [73]–[76]. A higher-order mode can be excited at the fiber discontinuity occurring at the first splice and then converted back to the fundamental mode at the second connector or splice. Since a higher-order mode cannot propagate far from its excitation point, this problem can be avoided by ensuring that the spacing between two connectors or splices exceeds 2 m. Generally speaking, modal noise is not a problem for properly designed and maintained single-mode fiber-optic communication systems.

With the development of the vertical-cavity surface-emitting laser (VCSEL), the modal-noise issue has resurfaced in recent years [77]–[81]. The use of such lasers in short-haul optical data links, making use of multimode fibers (even those made of plastic), is of considerable interest because of the high bandwidth associated with VCSELs. Indeed, rates of several gigabits per second have been demonstrated in laboratory experiments with plastic fibers [83]. However, VCSELs have a long coherence length as they oscillate in a single longitudinal mode. In a 1994 experiment the BER measure-

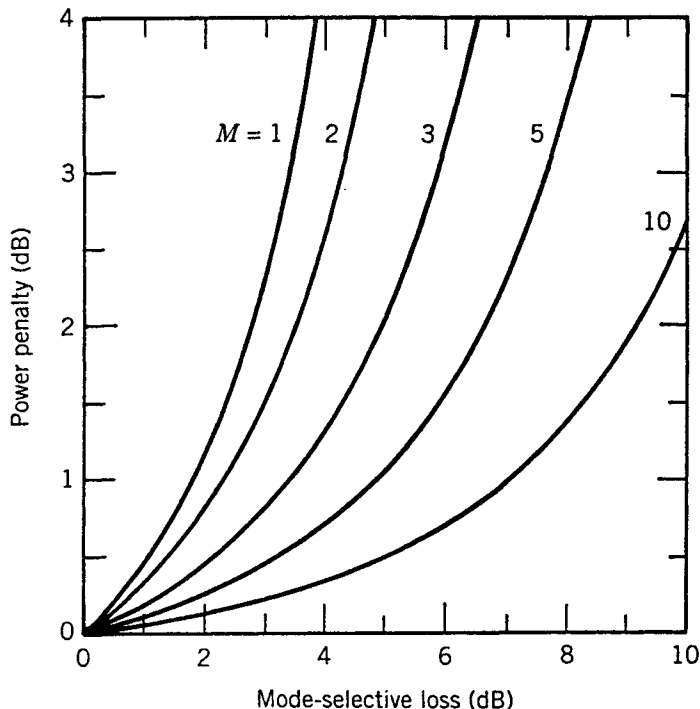


Figure 5.6: Modal-noise power penalty versus mode-selective loss. The parameter M is defined as the total number of longitudinal modes whose power exceeds 10% of the peak power. (After Ref. [71]; ©1986 IEEE; reprinted with permission.)

ments showed an error floor at a level of 10^{-7} even when the mode-selective loss was only 1 dB [78]. This problem can be avoided to some extent by using larger-diameter VCSELs which oscillate in several transverse modes and thus have a shorter coherence length. Computer models are generally used to estimate the power penalty for optical data links under realistic operating conditions [80]. Analytic tools such as the saddle-point method can also provide a reasonable estimate of the BER [81].

5.4.2 Mode-Partition Noise

As discussed in Section 3.3, multimode semiconductor lasers exhibit *mode-partition noise* (MPN), a phenomenon occurring because of an anticorrelation among pairs of longitudinal modes. In particular, various longitudinal modes fluctuate in such a way that individual modes exhibit large intensity fluctuations even though the total intensity remains relatively constant. MPN would be harmless in the absence of fiber dispersion, as all modes would remain synchronized during transmission and detection. In practice, different modes become unsynchronized, since they travel at slightly different speeds inside the fiber because of group-velocity dispersion. As a result of such desynchronization, the receiver current exhibits additional fluctuations, and the SNR at the

decision circuit becomes worse than that expected in the absence of MPN. A power penalty must be paid to improve the SNR to the same value that is necessary to achieve the required BER (see Section 4.5). The effect of MPN on system performance has been studied extensively for multimode semiconductor lasers [84]–[92].

In the case of multimode semiconductor lasers, the power penalty can be calculated by following an approach similar to that of Section 4.6.2 and is given by [84]

$$\delta_{\text{mpn}} = -5 \log_{10}(1 - Q^2 r_{\text{mpn}}^2), \quad (5.4.1)$$

where r_{mpn} is the relative noise level of the received power in the presence of MPN. A simple model for estimating the parameter r_{mpn} assumes that laser modes fluctuate in such a way that the total power remains constant under CW operation [84]. It also assumes that the average mode power is distributed according to a Gaussian distribution of RMS width σ_λ and that the pulse shape at the decision circuit of the receiver is described by a cosine function. Different laser modes are assumed to have the same cross-correlation coefficient γ_{cc} , i.e.,

$$\gamma_{\text{cc}} = \frac{\langle P_i P_j \rangle}{\langle P_i \rangle \langle P_j \rangle} \quad (5.4.2)$$

for all i and j such that $i \neq j$. The angular brackets denote an average over power fluctuations associated with mode partitioning. A straightforward calculation shows that r_{mpn} is given by [87]

$$r_{\text{mpn}} = (k/\sqrt{2}) \{1 - \exp[-(\pi BLD\sigma_\lambda)^2]\}, \quad (5.4.3)$$

where the mode-partition coefficient k is related to γ_{cc} as $k = \sqrt{1 - \gamma_{\text{cc}}}$. The model assumes that mode partition can be quantified in terms of a single parameter k with values in the range 0–1. The numerical value of k is difficult to estimate and is likely to vary from laser to laser. Experimental measurements suggest that the values of k are in the range 0.6–0.8 and vary for different mode pairs [89].

Equations (5.4.1) and (5.4.3) can be used to calculate the MPN-induced power penalty. Figure 5.7 shows the power penalty at a BER of 10^{-9} ($Q = 6$) as a function of the normalized dispersion parameter $BLD\sigma_\lambda$ for several values of the mode-partition coefficient k . For any value of k , δ_{mpn} increases rapidly with an increase in $BLD\sigma_\lambda$ and becomes infinite when $BLD\sigma_\lambda$ reaches a critical value. For $k > 0.5$, the MPN-induced power penalty becomes quite large if $BLD\sigma_\lambda$ exceeds 0.15. However, it can be reduced to a negligible level ($\delta_{\text{mpn}} < 0.5$ dB) by designing the optical communication system such that $BLD\sigma_\lambda < 0.1$. As an example, consider a 1.3- μm lightwave system. If we assume that the operating wavelength is matched to the zero-dispersion wavelength to within 10 nm, $D \approx 1$ ps/(km-nm). A typical value of σ_λ for multimode semiconductor lasers is 2 nm. The MPN-induced power penalty would be negligible if the BL product were below 50 (Gb/s)-km. At $B = 2$ Gb/s the transmission distance is then limited to 25 km. The situation becomes worse for 1.55- μm lightwave systems for which $D \approx 16$ ps/(km-nm) unless dispersion-shifted fibers are used. In general, the MPN-induced power penalty is quite sensitive to the spectral bandwidth of the multimode laser and can be reduced by reducing the bandwidth. In one study [92], a reduction in

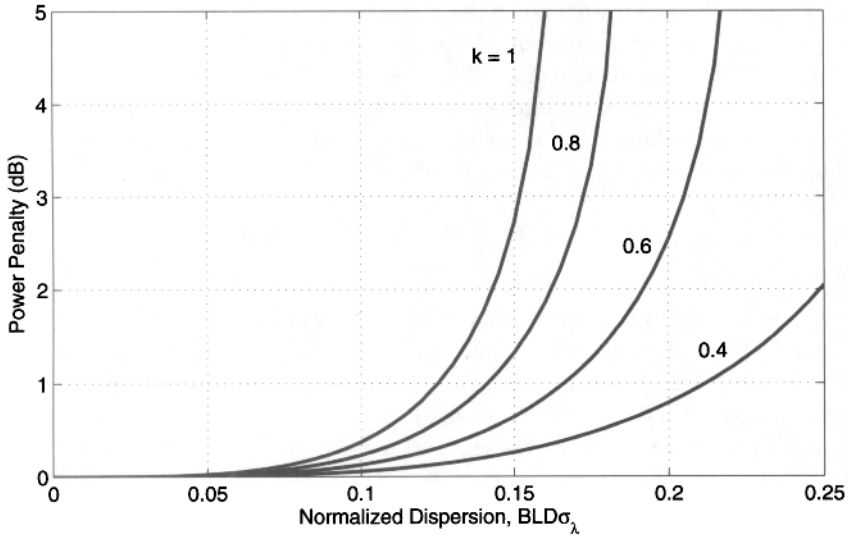


Figure 5.7: MPN-induced Power penalty versus $BLD\sigma_\lambda$ for a multimode semiconductor laser of RMS spectral width σ_λ . Different curves correspond to different values of the mode-partition coefficient k .

the carrier lifetime from 340 to 130 ps, realized by p -doping of the active layer, reduced the bandwidth of 1.3- μm semiconductor lasers by only 40% (from 5.6 to 3.4 nm), but the power penalty decreased from an infinite value (BER floor above 10^{-9} level) to a mere 0.5 dB.

5.4.3 Reflection Feedback and Noise

In most fiber-optic communication systems, some light is invariably reflected back because of refractive-index discontinuities occurring at splices, connectors, and fiber ends. The effects of such unintentional feedback have been studied extensively [93]–[104] because they can degrade the performance of lightwave systems considerably. Even a relatively small amount of optical feedback affects the operation of semiconductor lasers [97] and can lead to excess noise in the transmitter output. Even when an isolator is used between the transmitter and the fiber, multiple reflections between splices and connectors can generate additional intensity noise and degrade receiver performance [99]. This subsection is devoted to the effect of reflection-induced noise on receiver sensitivity.

Most reflections in a fiber link originate at glass–air interfaces whose reflectivity can be estimated by using $R_f = (n_f - 1)^2 / (n_f + 1)^2$, where n_f is the refractive index of the fiber material. For silica fibers $R_f = 3.6\%$ (–14.4 dB) if we use $n_f = 1.47$. This value increases to 5.3% for polished fiber ends since polishing can create a thin surface layer with a refractive index of about 1.6. In the case of multiple reflections occurring between two splices or connectors, the reflection feedback can increase con-

siderably because the two reflecting surfaces act as mirrors of a Fabry–Perot interferometer. When the resonance condition is satisfied, the reflectivity increases to 14% for unpolished surfaces and to over 22% for polished surfaces. Clearly, a considerable fraction of the signal transmitted can be reflected back unless precautions are taken to reduce the optical feedback. A common technique for reducing reflection feedback is to use index-matching oil or gel near glass–air interfaces. Sometimes the tip of the fiber is curved or cut at an angle so that the reflected light deviates from the fiber axis. Reflection feedback can be reduced to below 0.1% by such techniques.

Semiconductor lasers are extremely sensitive to optical feedback [101]; their operating characteristics can be affected by feedback as small as -80 dB [97]. The most dramatic effect of feedback is on the laser linewidth, which can narrow or broaden by several order of magnitude, depending on the exact location of the surface where feedback originates [93]. The reason behind such a sensitivity is related to the fact that the phase of the reflected light can perturb the laser phase significantly even for relatively weak feedback levels. Such feedback-induced phase changes are detrimental mainly for coherent communication systems. The performance of direct-detection lightwave systems is affected by intensity noise rather than phase noise.

Optical feedback can increase the intensity noise significantly. Several experiments have shown a feedback-induced enhancement of the intensity noise occurring at frequencies corresponding to multiples of the external-cavity mode spacing [94]–[96]. In fact, there are several mechanisms through which the relative intensity noise (RIN) of a semiconductor laser can be enhanced by the external optical feedback. In a simple model [98], the feedback-induced enhancement of the intensity noise is attributed to the onset of multiple, closely spaced, external-cavity longitudinal modes whose spacing is determined by the distance between the laser output facet and the glass–air interface where feedback originates. The number and the amplitudes of the external-cavity modes depend on the amount of feedback. In this model, the RIN enhancement is due to intensity fluctuations of the feedback-generated side modes. Another source of RIN enhancement has its origin in the feedback-induced chaos in semiconductor lasers. Numerical simulations of the rate equations show that the RIN can be enhanced by 20 dB or more when the feedback level exceeds a certain value [102]. Even though the feedback-induced chaos is deterministic in nature, it manifests as an apparent RIN increase.

Experimental measurements of the RIN and the BER in the presence of optical feedback confirm that the feedback-induced RIN enhancement leads to a power penalty in lightwave systems [105]–[107]. Figure 5.8 shows the results of the BER measurements for a VCSEL operating at 958 nm. Such a laser operates in a single longitudinal mode because of an ultrashort cavity length (~ 1 μ m) and exhibits a RIN near -130 dB/Hz in the absence of reflection feedback. However, the RIN increases by as much as 20 dB when the feedback exceeds the -30 -dB level. The BER measurements at a bit rate of 500 Mb/s show a power penalty of 0.8 dB at a BER of 10^{-9} for -30 -dB feedback, and the penalty increases rapidly at higher feedback levels [107].

The power penalty can be calculated by following the analysis of Section 4.6.2 and is given by

$$\delta_{\text{ref}} = -10 \log_{10}(1 - r_{\text{eff}}^2 Q^2), \quad (5.4.4)$$

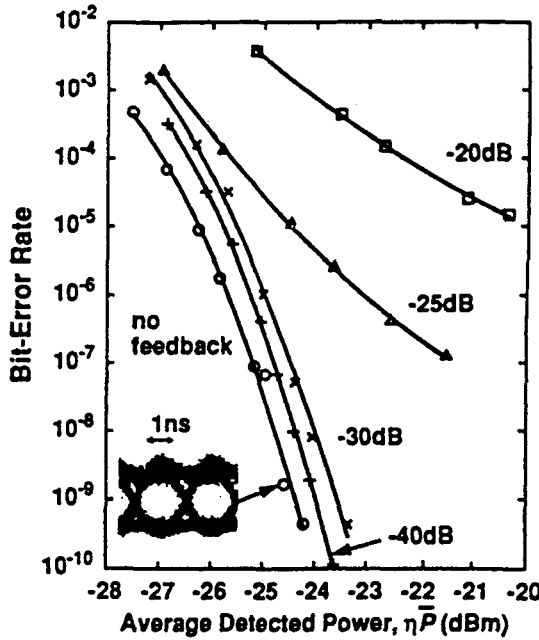


Figure 5.8: Experimentally measured BER at 500 Mb/s for a VCSEL under optical feedback. The BER is measured at several feedback levels. (After Ref. [107]; ©1993 IEEE; reprinted with permission.)

where r_{eff} is the effective intensity noise over the receiver bandwidth Δf and is obtained from

$$r_{\text{eff}}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{RIN}(\omega) d\omega = 2(\text{RIN})\Delta f. \quad (5.4.5)$$

In the case of feedback-induced external-cavity modes, r_{eff} can be calculated by using a simple model and is found to be [98]

$$r_{\text{eff}}^2 \approx r_I^2 + N/(\text{MSR})^2, \quad (5.4.6)$$

where r_I is the relative noise level in the absence of reflection feedback, N is the number of external-cavity modes, and MSR is the factor by which the external-cavity modes remain suppressed. Figure 5.9 shows the reflection-noise power penalty as a function of MSR for several values of N by choosing $r_I = 0.01$. The penalty is negligible in the absence of feedback ($N = 0$). However, it increases with an increase in N and a decrease in MSR. In fact, the penalty becomes infinite when MSR is reduced below a critical value. Thus, reflection feedback can degrade system performance to the extent that the system cannot achieve the desired BER despite an indefinite increase in the power received. Such reflection-induced BER floors have been observed experimentally [96] and indicate the severe impact of reflection noise on the performance of lightwave systems. An example of the reflection-induced BER floor is seen in Figure 5.8, where the BER remains above 10^{-9} for feedback levels in excess of -25 dB. Generally

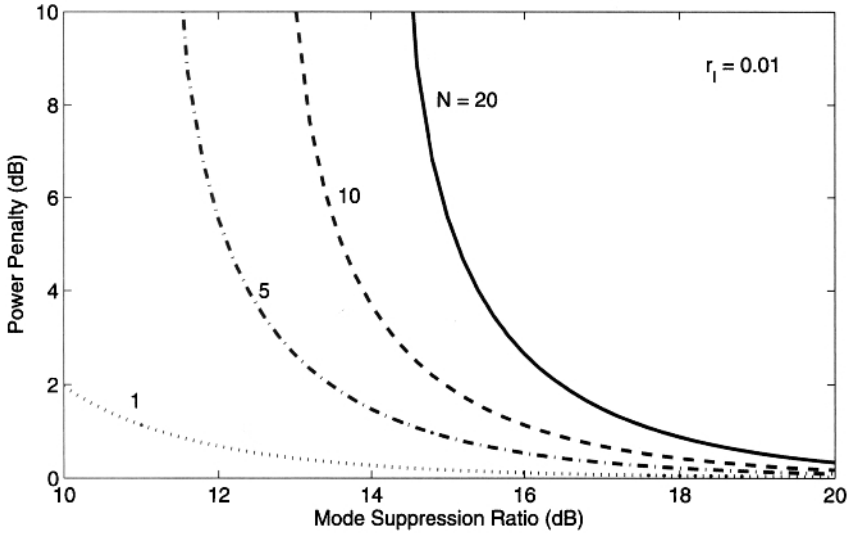


Figure 5.9: Feedback-induced power penalty as a function of MSR for several values of N and $r_l = 0.01$. Reflection feedback into the laser is assumed to generate N side modes of the same amplitude.

speaking, most lightwave systems operate satisfactorily when the reflection feedback is below -30 dB. In practice, the problem can be nearly eliminated by using an optical isolator within the transmitter module.

Even when an isolator is employed, reflection noise can be a problem for lightwave systems. In long-haul fiber links, fiber dispersion can convert phase noise into intensity noise, leading to performance degradation [100]. Similarly, two reflecting surfaces anywhere along the fiber link act as a Fabry–Perot interferometer which can convert phase noise into intensity noise [99]. Such a conversion can be understood by noting that multiple reflections inside a Fabry–Perot interferometer lead to a phase-dependent term in the transmitted intensity which fluctuates in response to phase fluctuations. As a result, the RIN of the signal incident on the receiver is higher than that occurring in the absence of reflection feedback. Most of the RIN enhancement occurs over a narrow frequency band whose spectral width is governed by the laser linewidth (~ 100 MHz). Since total noise is obtained by integrating over the receiver bandwidth, it can affect system performance considerably at bit rates larger than 1 Gb/s. The power penalty can still be calculated by using Eq. (5.4.4). A simple model that includes only two reflections between the reflecting interfaces shows that r_{eff} is proportional to $(R_1 R_2)^{1/2}$, where R_1 and R_2 are the reflectivities of the two interfaces [99]. Figure 4.21 can be used to estimate the power penalty. It shows that power penalty can become infinite and lead to BER floors when r_{eff} exceeds 0.2. Such BER floors have been observed experimentally [99]. They can be avoided only by eliminating or reducing parasitic reflections along the entire fiber link. It is therefore necessary to employ connectors and splices that reduce reflections through the use of index matching or other techniques.

5.4.4 Dispersive Pulse Broadening

Dispersion-induced pulse broadening affects the receiver performance in two ways. First, a part of the pulse energy spreads beyond the allocated bit slot and leads to inter-symbol interference. Second, the pulse energy within the bit slot is reduced when the optical pulse broadens. Such a decrease in pulse energy reduces the SNR at the decision circuit. As the SNR should remain constant to maintain the system performance, the receiver requires more average power. This is the origin of dispersion-induced power penalty δ_d .

An exact calculation of δ_d is difficult, as it depends on many details such as the extent of pulse shaping at the receiver. A rough estimate is obtained by following the analysis of Section 3.3.1, where broadening of Gaussian pulses is discussed. Equation (3.3.9) shows that the optical pulse remains Gaussian, but its peak power is reduced by a pulse-broadening factor given in Eq. (3.3.10). If we define the power penalty δ_d as the increase (in decibels) in the received power that would compensate the peak-power reduction, δ_d is given by

$$\delta_d = 10 \log_{10} b_f, \quad (5.4.7)$$

where b_f is the pulse broadening factor. As in Section 3.3.4, we consider the case of broadband and narrowband optical sources separately.

Consider first a lightwave system designed with a relatively broadband optical source. The broadening factor b_f in this case is obtained from Eq. (2.4.24) and has the form

$$b_f = \sigma/\sigma_0 = [1 + (DL\sigma_\lambda/\sigma_0)^2]^{1/2}, \quad (5.4.8)$$

where σ_λ is the RMS width of the source spectrum. The RMS width σ_0 of the optical pulse at the transmitter end is a design parameter. It can be related to the duty cycle d_c of RZ pulses as $4\sigma_0 = d_c T_b$, where $T_b \equiv 1/B$ is the duration of the bit slot at a given bit rate B . Using $\sigma_0 = d_c/(4B)$ in Eq. (5.4.8) and using Eq. (5.4.7), the power penalty becomes

$$\delta_d = 5 \log_{10} [1 + (4BLD\sigma_\lambda/d_c)^2]. \quad (5.4.9)$$

This result should be compared with the condition (3.3.40) found in Section 3.3.4. If we assume that input pulses are wide enough to occupy the entire bit slot ($d_c = 1$), the power penalty is negligible for $4BLD\sigma_\lambda \ll 1$, becomes 1.5 dB when $4BLD\sigma_\lambda = 1$, and increases rapidly beyond that.

In the case of a long-haul system designed with a narrowband source and chirp-free pulses, the broadening factor is obtained from Eq. (2.4.29). If we use again $\sigma_0 = d_c/(4B)$, the power penalty is given by

$$\delta_d = 5 \log_{10} [1 + (8\beta_2 B^2 L/d_c^2)^2]. \quad (5.4.10)$$

Figure 5.10 shows power penalty as a function of the dimensionless parameter combination $\mu = |\beta_2|B^2L$ for three values of d_c . Although the power penalty is negligible for values of $\mu < 0.05$ and $d_c > 0.5$, it increases rapidly as μ increases and exceeds 5 dB for $\mu = 0.1$ and $d_c = 0.5$. It is thus important to keep μ below 0.1. As an example, the operating distance of a 10-Gb/s lightwave system is limited to below 50 km because of dispersion when standard fibers are used with $|\beta_2| \approx 20 \text{ ps}^2/\text{km}$, but this value can be

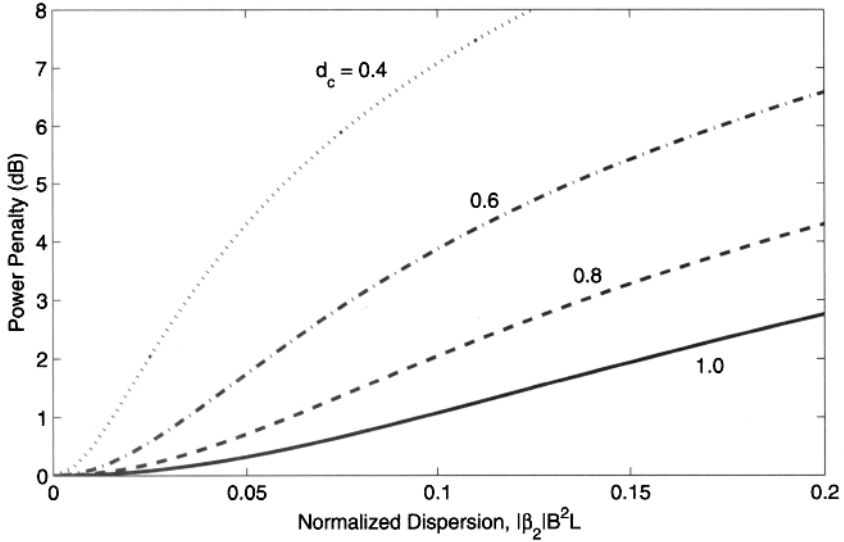


Figure 5.10: Dispersion-induced power penalty function as a function of $\mu = |\beta_2|B^2L$ for three values of duty cycle associated with an RZ bit stream.

increased considerably through dispersion management. It should be stressed that Eq. (5.4.10) provides only a rough estimate as its derivation is based on the assumptions of a Gaussian pulse shape.

5.4.5 Frequency Chirping

The preceding discussion of dispersion-induced power penalty assumed that the input pulses were unchirped. An initial chirp on optical pulses is known to limit the performance of 1.55- μm lightwave systems when directly modulated semiconductor lasers are used to generate the digital bit stream [108]–[121]. As discussed in Section 2.4.2, frequency chirping can enhance the dispersion-induced broadening of optical pulses and thus may degrade the performance of a long-haul lightwave system more than that expected when unchirped pulses are employed.

An exact calculation of the chirp-induced power penalty δ_c is difficult because frequency chirp depends on both the shape and the width of the optical pulse [110]–[113]. However, if we assume a Gaussian pulse shape and a linear chirp, the analysis of Section 2.4.2 can be used to estimate the chirp-induced power penalty. If we use Eq. (2.4.17) for the pulse broadening factor in Eq. (5.4.7) together with $T_0 = \sqrt{2}d_c/(4B)$, the power penalty is given by

$$\delta_c = 5 \log_{10} [(1 + 8C\beta_2 B^2 L / d_c^2)^2 + (8\beta_2 B^2 L / d_c^2)^2]. \quad (5.4.11)$$

Figure 5.11 shows the chirp-induced power penalty as a function of $|\beta_2|B^2L$ for several values of the chirp parameter C with $d_c = 1$. The parameter β_2 is taken to be

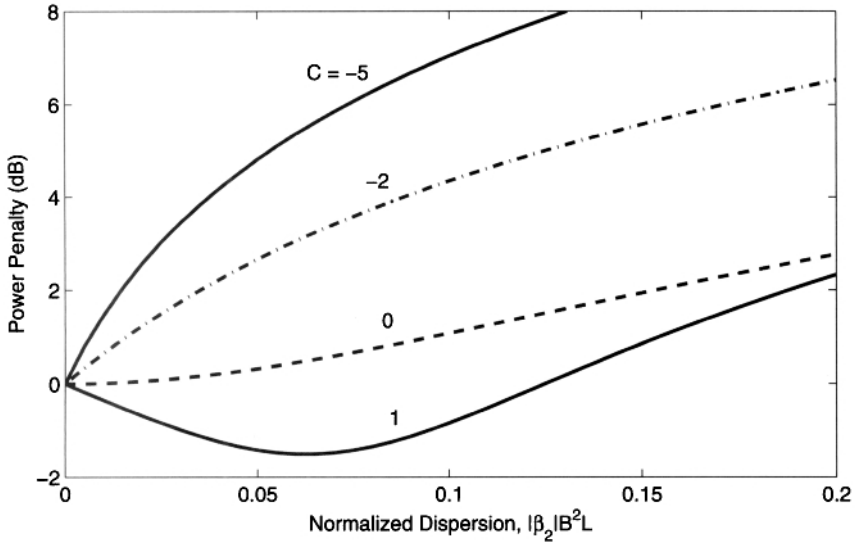


Figure 5.11: Chirp-induced power penalty as a function of $|\beta_2|B^2L$ for several values of the chirp parameter C . Gaussian pulses are assumed to be linearly chirped.

negative, as is the case for $1.55\text{-}\mu\text{m}$ lightwave systems. The $C = 0$ curve corresponds to the case of a chirp-free pulse. Power penalty is negligible (<0.1 dB) in this ideal case as long as $|\beta_2|B^2L < 0.05$. However, the penalty can exceed 5 dB if the transmitted pulses are chirped such that $C = -6$ (a typical value for semiconductor lasers). To keep the penalty below 0.1 dB, the system should be designed with $|\beta_2|B^2L < 0.002$. For standard optical fibers with $\beta_2 \approx -20 \text{ ps}^2/\text{km}$, B^2L is limited to $100 (\text{Gb/s})^2\text{-km}$, indicating that even at $B = 2.5 \text{ Gb/s}$, the transmission distance is limited to below 16 km because of frequency chirping. Interestingly, system performance can be improved by ensuring that $\beta_2 C < 0$. As discussed in Section 3.3, each optical pulse then undergoes an initial compression phase. As C is negative for semiconductor lasers, fibers with “normal” dispersion ($\beta_2 > 0$) can provide better performance whenever directly modulated semiconductor lasers are employed. For this reason, fibers with normal GVD are often employed in metropolitan area networks. Alternatively, one can make use of dispersion compensation and ensure that the average value of β_2 is close to zero.

5.4.6 Eye-Closure Penalty

An alternative measure of system performance is provided by the extent the “eye opening” in the eye diagram is affected by the dispersive and nonlinear effects accumulated inside the fiber link. As discussed in Section 4.3, an electrical filter with a bandwidth smaller than the bit rate is used inside the receiver to reshape pulses before they reach the decision circuit. Figure 4.14 shows the eye diagram looks like for an NRZ-format bit stream. When RZ format is employed, the pattern still has the appearance of an eye, even though the top horizontal rail is missing. Even in the case of the DPSK format,

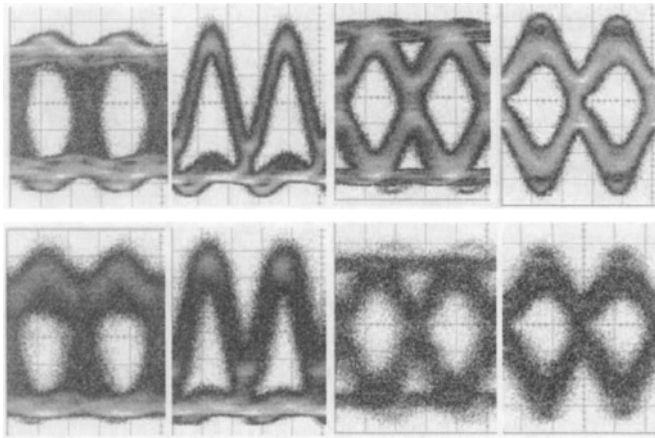


Figure 5.12: Measured eye diagrams at 40 Gb/s under back-to-back conditions (top row) and after a 263-km-long fiber link (bottom row). In both rows successive traces corresponds to NRZ, RZ, NRZ-DPSK, and RZ-DPSK formats, respectively. (After Ref. [122]; ©2004 IEEE.)

the eye diagram retains its shape. The top row in Figure 5.12 shows the measured eye diagrams at 40 Gb/s in the case of NRZ, CSRZ, NRZ-DPSK, and RZ-DPSK formats under the back-to-back condition in which the transmitter is directly connected to the receiver.

As the optical bit stream is transmitted through a fiber link, the accumulation of dispersive and nonlinear effects distorts optical pulses. These distortions manifest in the eye diagram through a reduced opening of the eye. The bottom row in Figure 5.12 shows the measured eye diagrams at 40 Gb/s after a 263-km-long fiber link for the same four modulation formats [122]. As seen there, eye opening is reduced for all formats. Since the decision threshold is set in the center of the open portion of the eye, any reduction in eye opening indicates an increase in the BER. This observation relates eye closure to the BER and suggests that its magnitude can provide a measure of the system performance. More precisely, the eye-closure penalty is quantified (in dB) as

$$\delta_{\text{eye}} = -10 \log_{10} \left(\frac{\text{eye opening after transmission}}{\text{eye opening before transmission}} \right). \quad (5.4.12)$$

Before Eq. (5.4.12) can be used, we need to clarify what one means by eye opening. Ideally, the eye amplitude peaks in the center of the bit slot and it provides an appropriate measure of eye opening. However, in practice, timing jitter makes it difficult to sample each pulse precisely at the location where pulse amplitude is maximum. If we allow an uncertainty of up to 10% on each side of the decision threshold, we should consider a rectangle of the largest area with a base of $0.2T_b$, where T_b is the duration of each symbol, that can be fitted within the open portion of the eye. The height of this rectangle then measures eye opening. This approach is commonly adopted for numerical simulations.

5.5 Forward Error Correction

As seen in the preceding section, receiver sensitivity and the BER of a lightwave system are degraded by many factors that are not always controllable in practice. Depending on details of the system design and objectives, it is entirely possible that a specified BER cannot be achieved. Under such conditions, the use of an error-correction scheme remains the only viable alternative.

Error control is not a new concept and is employed widely in electrical systems dealing with the transfer of digital data from one device to another [123]–[126]. The techniques used for controlling errors can be divided into two groups. In one group, errors are detected but not corrected. Rather, each packet of bits received with errors is retransmitted. This approach is suitable when data bits are transmitted in the form of packets (as is the case for the protocol used by the Internet) and they do not arrive at the destination in a synchronous fashion. In the other group, errors are detected as well as corrected at the receiver end without any retransmission of bits. This approach is referred to as forward error correction (FEC) and is best suited for lightwave systems operating with a synchronous protocol such as SONET or SDH.

Historically, lightwave systems did not employ FEC until the use of in-line optical amplifiers became common [127]–[129]. The use of FEC accelerated with the advent of WDM technology. As early as 1996, FEC was employed for a WDM system designed to operate over more than 425 km without any in-line amplifier or regenerator [130]. Since then, the FEC technique has been used in many WDM systems and is now considered almost routine [131]–[134].

5.5.1 Error-Correcting Codes

The basic idea behind any error-control technique is to add extra bits to the signal at the transmitter end in a judicious manner using a suitable coding algorithm [123]–[126]. A simple example is provided by the so-called parity bit that is added to the 7-bit ASCII code. In this example, the parity bit is chosen to be 0 or 1 depending on whether the number of 1 bits in the 7-bit sequence is even or odd. If a single bit is in error at the receiving end, an examination of the parity bit reveals the error.

The situation is somewhat different in the case of an optical bit stream, but the basic idea remains the same. An encoder within the transmitter adds additional control bits using a suitable code. At the receiver end, a decoder uses these control bits to detect errors, while correcting them simultaneously. How many errors can be corrected depends on the coding scheme employed. In general, more errors can be corrected by adding more control bits to the signal. Clearly, there is a limit to this process since the bit rate of the signal increases after the decoder. If B_e is the effective bit rate after coding a signal at the bit rate B , the *FEC overhead* associated with the error-correcting code is $B_e/B - 1$. The concept of *redundancy* is also used for FEC codes as the bits added by the coding scheme do not carry any information. Redundancy of a code is defined as $\rho = 1 - B/B_e$.

Many different types of error-correcting codes have been developed, often classified under names such as linear, cyclic, Hamming, Reed–Solomon, convolutional, product, and turbo codes [131]. Among these, Reed–Solomon (RS) codes have at-

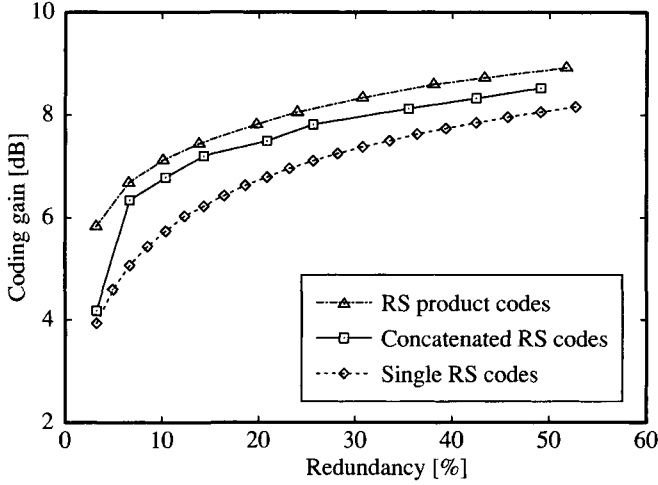


Figure 5.13: Coding gain as a function of code redundancy (overhead) for single, concatenated, and product-type RS codes. (After Ref. [132]; ©2002 IEEE.)

tracted most attention in the context of lightwave systems [132]. An RS code is denoted as $RS(n, k)$, where k is the size of a packet of bits that is converted through coding into a larger packet with n bits. The value of n is chosen such that $n = 2^m - 1$, where m is an integer. The RS code recommended by ITU for submarine applications uses $m = 8$ and is written as $RS(255, 239)$. The FEC overhead for this code is only 6.7%. Many other RS codes can be used if a higher overhead is permitted. For example, the code $RS(255, 207)$ has an overhead of 23.2% but it allows for more robust error control. The choice of the code depends on the level of improvement in the BER required for the system to operate reliably. It is common to quantify this improvement through the *coding gain*, a concept we discuss next.

5.5.2 Coding Gain

Coding gain is a measure of the improvement in BER realized through FEC. Since BER is related to the Q factor as indicated in Eq. (4.6.10), it is often expressed in terms of the equivalent value of Q that corresponds to the BER realized after the FEC decoder. The coding gain in decibel units is defined as [132]

$$G_c = 20 \log_{10}(Q_c/Q), \quad (5.5.1)$$

where Q_c and Q are related to the BERs obtained with and without FEC as

$$BER_c = \frac{1}{2} \operatorname{erfc}(Q_c/\sqrt{2}), \quad BER = \frac{1}{2} \operatorname{erfc}(Q/\sqrt{2}). \quad (5.5.2)$$

The factor of 20 appears in Eq. (5.5.1) in place of 10 because Q^2 is traditionally used for expressing the Q factor in decibel units. As an example, if the FEC decoder improves the BER from its original value of 10^{-3} to 10^{-9} , the value of Q increases from about 3

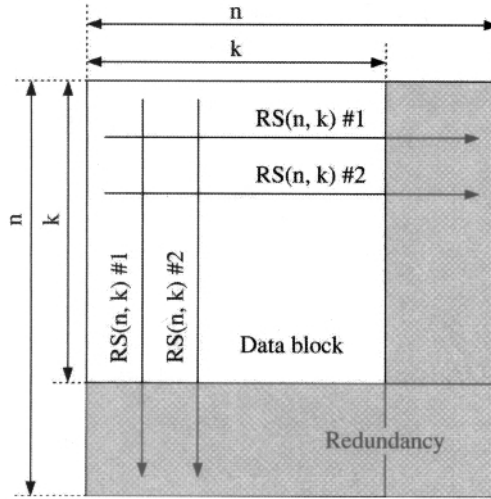


Figure 5.14: Schematic illustration of a RS product code. The same code is applied along the rows and columns of a block of bits. (After Ref. [132]; ©2002 IEEE.)

to 6, resulting in a coding gain of 6 dB. The coding gain is sometimes defined in terms of the SNR [131]. The two definitions differ by a small amount of $10 \log_{10}(B_e/B)$.

As one would expect, the magnitude of coding gain increases with the FEC overhead (or redundancy). The dashed line in Figure 5.13 shows this behavior. The coding gain is about 5.5 dB for 10% overhead and increases sublinearly, reaching only 8 dB even for a 50% overhead. It can be improved by concatenating two or more RS codes or by employing the RS product codes, but in all cases the coding gain begins to saturate as overhead increases. In the case of a RS product code, more than 6 dB of coding gain can be realized with only 5% overhead. The basic idea behind an RS product code is shown in Figure 5.14. As seen there, a block of data with k^2 bits is converted into n^2 bits by applying the same $RS(n, k)$ code both along the rows and columns. As a result, the overhead of $n^2/k^2 - 1$ for a RS product code is larger, but it also allows more error control.

5.6 Computer-Aided Design

The design of a fiber-optic communication system involves optimization of a large number of parameters associated with transmitters, optical fibers, in-line amplifiers, and receivers. The design aspects discussed in Section 5.2 are too simple to provide the optimized values for all system parameters. The power and the rise-time budgets are only useful for obtaining a conservative estimate of the transmission distance (or repeater spacing) and the bit rate. The system margin in Eq. (5.2.2) is used as a vehicle to include various sources of power penalties discussed in Section 5.4. Such a simple approach fails for modern high-capacity systems designed to operate over long distances using optical amplifiers.

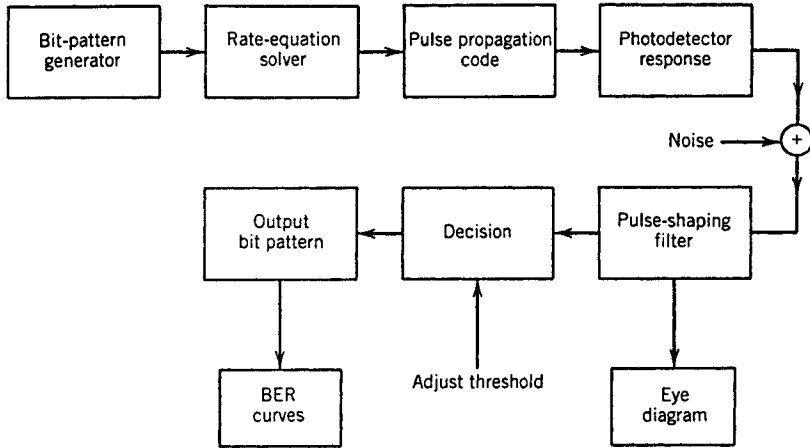


Figure 5.15: Steps involved in computer modeling of fiber-optic communication systems.

An alternative approach uses computer simulations and provides a much more realistic modeling of fiber-optic communication systems [136]–[149]. The computer-aided design techniques are capable of optimizing the whole system and can provide the optimum values of various system parameters such that the design objectives are met at a minimum cost. Figure 5.15 illustrates the various steps involved in the simulation process. The approach consists of generating an optical bit pattern at the transmitter, transmitting it through the fiber link, detecting it at the receiver, and then analyzing it through the tools such as the eye diagram and the Q factor.

Each step in the block diagram shown in Figure 5.15 can be carried out numerically by using the material given in Chapters 2–4. The input to the optical transmitter is a pseudorandom sequence of electrical pulses which represent 1 and 0 bits. The length N of the pseudorandom bit sequence determines the computing time and should be chosen judiciously. Typically, $N = 2^M$ with M is in the range of 6 to 10. The optical bit stream can be obtained by solving the rate equations that govern the modulation response of semiconductor lasers (see Section 3.3). The equations governing the modulation response should be used if an external modulator is used. Chirping is automatically included in both cases.

The most time-consuming part of system simulations is propagation of the optical bit stream through the fiber link which may contain multiple fiber sections with optical amplifiers inserted in between. Changes to the bit stream occurring in each fiber section are calculated by solving the NLS equation (5.3.1) so that both the dispersive and nonlinear effects are fully included. The noise added by optical amplifiers should also be included at the location of each amplifier.

The optical bit stream is converted into the electrical domain at the photodetector where shot and thermal noises are added as discussed in Section 4.4. The electrical bit stream is then passed through a pulse-shaping filter whose bandwidth is also a design parameter. An eye diagram is constructed using this filtered bit stream. The effect of varying system parameters can be studied by monitoring the eye degradation or by

calculating the Q parameter given in Eq. (4.6.11). Such an approach can be used to obtain the power penalty associated with various mechanisms discussed in Section 5.4. It can also be used to investigate trade-offs that would optimize the overall system performance. Numerical simulations reveal the existence of an optimum extinction ratio for which the system penalty is minimum.

Computer-aided design has another important role to play. A long-haul lightwave system may contain many repeaters, both optical and electrical. Transmitters, receivers, and amplifiers used at repeaters, although chosen to satisfy nominal specifications, are never identical. Similarly, fiber cables are constructed by splicing many different pieces (typical length 4–8 km) which have slightly different loss and dispersion characteristics. The net result is that many system parameters vary around their nominal values. For example, the dispersion parameter D , responsible not only for pulse broadening but also for other sources of power penalty, can vary significantly in different sections of the fiber link because of variations in the zero-dispersion wavelength and the transmitter wavelength. A statistical approach is often used to estimate the effect of such inherent variations on the performance of a realistic lightwave system. The idea behind such an approach is that it is extremely unlikely that all system parameters would take their worst-case values at the same time. Thus, repeater spacing can be increased well above its worst-case value if the system is designed to operate reliably at the specific bit rate with a high probability (say 99.9%).

The importance of computer-aided design for fiber-optic communication systems became apparent during the 1990s when the dispersive and nonlinear effects in optical fibers became of paramount concern with increasing bit rates and transmission distances. All modern lightwave systems are designed using numerical simulations, and several software packages are available commercially. Appendix D provides details on the simulation package available on the CD included with this book (courtesy of OptiWave Corporation). The CD also contains many additional problems designed specifically for a better understanding of the material covered in this book. The reader is encouraged to use the software on this CD to develop additional skills.

Problems

- 5.1** A distribution network uses an optical bus to distribute the signal to 10 users. Each optical tap couples 10% of the power to the user and has 1-dB insertion loss. Assuming that the station 1 transmits 1 mW of power over the optical bus, calculate the power received by the stations 8, 9, and 10.
- 5.2** A cable-television operator uses an optical bus to distribute the video signal to its subscribers. Each receiver needs a minimum of 100 nW to operate satisfactorily. Optical taps couple 5% of the power to each subscriber. Assuming 0.5 dB insertion loss for each tap and 1 mW transmitter power, estimate the number of subscribers that can be added to the optical bus?
- 5.3** A star network uses directional couplers with 0.5-dB insertion loss to distribute data to its subscribers. If each receiver requires a minimum of 100 nW and each

transmitter is capable of emitting 0.5 mW, calculate the maximum number of subscribers served by the network.

- 5.4** Make the power budget and calculate the maximum transmission distance for a 1.3- μm lightwave system operating at 100 Mb/s and using an LED for launching 0.1 mW of average power into the fiber. Assume 1-dB/km fiber loss, 0.2-dB splice loss every 2 km, 1-dB connector loss at each end of fiber link, and 100-nW receiver sensitivity. Allow 6-dB system margin.
- 5.5** A 1.3- μm long-haul lightwave system is designed to operate at 1.5 Gb/s. It is capable of coupling 1 mW of average power into the fiber. The 0.5-dB/km fiber-cable loss includes splice losses. The connectors at each end have 1-dB losses. The InGaAs $p-i-n$ receiver has a sensitivity of 250 nW. Make the power budget and estimate the repeater spacing.
- 5.6** Prove that the rise time T_r and the 3-dB bandwidth Δf of a RC circuit are related by $T_r \Delta f = 0.35$.
- 5.7** Consider a super-Gaussian optical pulse with the power distribution

$$P(t) = P_0 \exp[-(t/T_0)^{2m}],$$

where the parameter m controls the pulse shape. Derive an expression for the rise time T_r of such a pulse. Calculate the ratio T_r/T_{FWHM} , where T_{FWHM} is the full width at half maximum, and show that for a Gaussian pulse ($m = 1$) this ratio equals 0.716.

- 5.8** Prove that for a Gaussian optical pulse, the rise time T_r and the 3-dB optical bandwidth Δf are related by $T_r \Delta f = 0.316$.
- 5.9** Make the rise-time budget for a 0.85- μm , 10-km fiber link designed to operate at 50 Mb/s. The LED transmitter and the Si $p-i-n$ receiver have rise times of 10 and 15 ns, respectively. The graded-index fiber has a core index of 1.46, $\Delta = 0.01$, and $D = 80 \text{ ps}/(\text{km}\cdot\text{nm})$. The LED spectral width is 50 nm. Can the system be designed to operate with the NRZ format?
- 5.10** A 1.3- μm lightwave system is designed to operate at 1.7 Gb/s with a repeater spacing of 45 km. The single-mode fiber has a dispersion slope of 0.1 ps/(km·nm²) in the vicinity of the zero-dispersion wavelength occurring at 1.308 μm . Calculate the wavelength range of multimode semiconductor lasers for which the mode-partition-noise power penalty remains below 1 dB. Assume that the RMS spectral width of the laser is 2 nm and the mode-partition coefficient $k = 0.7$.
- 5.11** Generalize Eq. (5.4.1) for the case of APD receivers by including the excess-noise factor in the form $F(M) = M^x$.
- 5.12** Consider a 1.55- μm lightwave system operating at 1 Gb/s by using multimode semiconductor lasers of 2 nm (RMS) spectral width. Calculate the maximum transmission distance that would keep the mode-partition-noise power penalty below 2 dB. Use $k = 0.8$ for the mode-partition coefficient.
- 5.13** Use Eq. (5.4.11) to determine the maximum transmission distance for a 1.55- μm lightwave system operating at 4 Gb/s such that the chirp-induced power penalty

is below 1 dB. Assume that $C = -6$ for the single-mode semiconductor laser and $\beta_2 = -20 \text{ ps}^2/\text{km}$ for the single-mode fiber.

- 5.14** Repeat the preceding problem in the case of 8-Gb/s bit rate.
- 5.15** Use the results of Problem 4.18 to obtain an expression of the reflection-induced power penalty in the case of a finite extinction ratio r_{ex} . Reproduce the penalty curves shown in Figure 5.8 for the case $r_{\text{ex}} = 0.1$.
- 5.16** Consider a Fabry–Perot interferometer with two surfaces of reflectivity R_1 and R_2 . Follow the analysis of Ref. [99] to derive an expression of the relative intensity noise $\text{RIN}(\omega)$ of the transmitted light as a function of the linewidth of the incident light. Assume that R_1 and R_2 are small enough that it is enough to consider only a single reflection at each surface.
- 5.17** Follow the analysis of Ref. [136] to obtain an expression for the total receiver noise by including thermal noise, shot noise, intensity noise, mode-partition noise, chirp noise, and reflection noise.

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